Highly Efficient, Regulated Dual-Output, Ambient Energy Manager for AC or DC Sources with Optional Primary Battery

Features

Maximum Power Point Tracking

- Open-circuit voltage sensing for MPPT every 0.33s.
- Selectable MPPT ratio to 50%, 65% or 80%.
- Input voltage up to 5 V with MPPT regulation from 50 mV to 4.5 mV.
- Constant impedance matching (ZMPPT).

Ultra-low power start-up

- Cold start from 380 mV input voltage and 3 µW input power (typical).

Integrated low-voltage LDO regulator

- Up to 20 mA load current.
- Output voltage configurable at 1.2 V or 1.8 V.
- Dynamically power-gated by external control.

Integrated high-voltage LDO regulator

- Up to 80 mA load current with 300 mV drop-out.
- Output voltage configurable within a range of 1.8 V to 4.1 V.
- Dynamically power-gated by external control.

Flexible energy storage management

- Configurable overcharge and over-discharge protection for any type of rechargeable storage elements.
- Fast supercapacitor charging.
- Status signal when storage element is running low and when LDO outputs are available.

Optional primary battery

- Automatic switching to primary battery when the storage element is depleted.

Integrated storage element balancing circuit for dual-cell supercapacitor

Applications

- Smart home
- Smart building
- Edge IoT
- Industrial sensor
- Retail
- PC accessories

Description

The AEM30940 is an integrated energy management circuit that extracts DC power from a piezo generator, a micro turbine generator or any high frequency RF input to simultaneously store energy in a rechargeable element through its boost converter and supply the system with two independent regulated output voltages. The AEM30940 allows to extend battery lifetime and eliminate the primary energy storage element in a large range of applications.

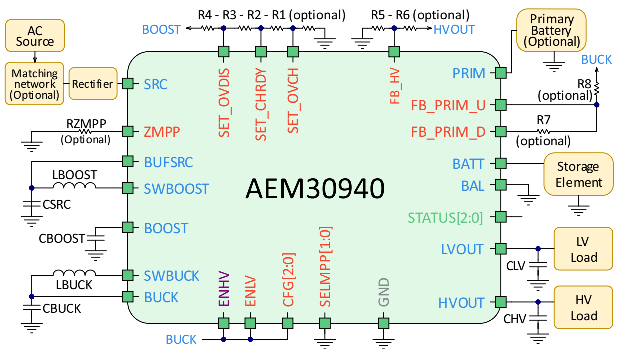
With its unique cold-start circuit, it can start operating from 380 mV and only 3 µW from the input with empty storage elements.

The low-voltage output typically drives a microcontroller while the high-voltage output typically drives a radio transceiver. Both are driven by highly-efficient LDO (Low Drop-Out) linear regulators for low noise and high stability.

Five capacitors and two inductors are required, all available in small packages. With only seven external components, integration is maximized, footprint and BOM are minimized, optimizing the time-to-market and design costs.

Device Information

Description	Part number
AEM30940 - QFN 28-pin 5x5mm	10AEM30940C0000
AEM30940 - Evaluation board	2AAEM30940C0x1



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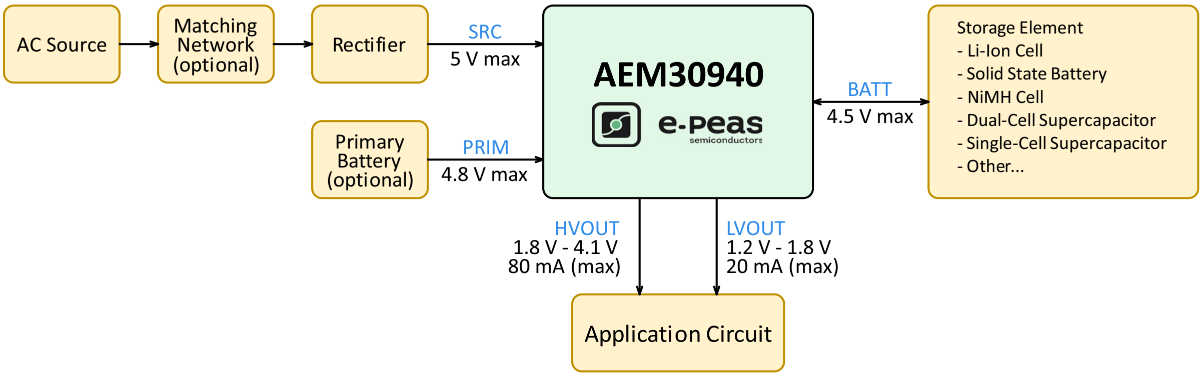


Figure 1: Simplified schematic view

1. Introduction

The AEM30940 is a full-featured energy efficient power management circuit capable of charging a storage element (battery or supercapacitor, connected to BATT) from an energy source (connected to SRC) as well as supplying loads at different operating voltages through two power supplying LDO regulators (LVOUT and HVOUT).

The heart of the AEM30940 is a cascade of two regulated switching converters, namely the boost converter and the buck converter, both with high power conversion efficiencies (See Section 3.6).

At first start-up, as soon as a required cold-start voltage of 380 mV and a scant amount of power of only 3 µW are available from the harvested energy source, the AEM coldstarts. After the cold start, the AEM can extract the power available from the source as long as the input voltage is within 50 mV to 5 V range.

Through three configuration pins (CFG[2:0]), the user can select a specific operating mode from a range of seven modes that covers most application requirements without any dedicated external component. These operating modes define the LDO output voltages and the storage element protection thresholds. A custom mode allows the user to define arbitrary storage element protection levels and the output voltage of the high-voltage LDO (See Section 6.1).

The Maximum Power Point Tracking (MPPT) ratio can be configured using two configuration pins (SELMPP[1:0]) (See Section 6.2).

Two logic control pins (ENLV and ENHV) allow to dynamically activate or deactivate the LDO regulators that supply the low and high voltage loads. The status pin STATUS[0] alerts the user that the LDOs are operational and can be enabled. This signal can also be used to enable an optional external regulator.

If the storage element voltage gets depleted, LVOUT and HVOUT are power-gated and the controller is no longer supplied by the storage element to protect it from further discharge. Around 600 ms before the shutdown of the AEM, the status pin STATUS[1] alerts the user for a clean shutdown of the system.

However, if the storage element gets depleted and an optional primary battery is connected to PRIM, the AEM30940 automatically uses it as a source to recharge the storage element before switching back to the ambient source connected to SRC. This guarantees continuous operation even under the most adverse conditions (See Section 5.2.4). STATUS[1] is asserted when the primary battery is providing power.

The status of the MPPT controller is reported with one dedicated status pin (STATUS[2]). The status pin is asserted when an MPP evaluation is being performed.

2. Pin Configuration and Functions

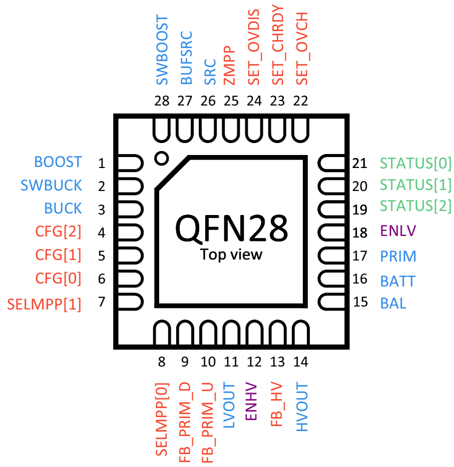


Figure 2: Pinout diagram QFN 28-pin

Name	Pin	Function
Power pins		
BOOST	1	Output of the boost converter.
SWBUCK	2	Switching node of the buck converter.
BUCK	3	Output of the buck converter.
LVOUT	11	Output of the low voltage LDO regulator.
HVOUT	14	Output of the high voltage LDO regulator.
BAL	15	Connection to the mid-point of a dual-cell supercapacitor (optional). Must be connected to GND if not used.
BATT	16	Connection to the energy storage element, battery or capacitor. Cannot be left floating.
PRIM	17	Connection to the primary battery (optional). Must be connected to GND if not used.
SRC	26	Connection to the harvested energy source.
BUFSRC	27	Connection to an external capacitor buffering the boost converter input.
SWBOOST	28	Switching node of the boost converter.

Table 1: Pins description (part 1)

<table>
<tr><th rowspan="2">Name</th><th rowspan="2">Pin</th><th colspan="2">Logic Level</th><th rowspan="2">Function</th></tr>
<tr><th>LOW</th><th>HIGH</th></tr>
<tr><td colspan="5">Configuration pins</td></tr>
<tr><td>CFG[2]</td><td>4</td><td>GND</td><td>BUCK</td><td rowspan="3">Used to configure the storage element protection thresholds and the LDOs output voltages.</td></tr>
<tr><td>CFG[1]</td><td>5</td><td>GND</td><td>BUCK</td></tr>
<tr><td>CFG[0]</td><td>6</td><td>GND</td><td>BUCK</td></tr>
<tr><td>SELMPP[1]</td><td>7</td><td>GND</td><td>BUCK</td><td rowspan="2">Used to configure the MPPT ratio.</td></tr>
<tr><td>SELMPP[0]</td><td>8</td><td>GND</td><td>BUCK</td></tr>
<tr><td>FB_PRIM_D</td><td>9</td><td colspan="2">Analog Pin</td><td rowspan="2">Used to configure the primary battery overdischarge voltage (optional).
Must be connected to GND if not used.</td></tr>
<tr><td>FB_PRIM_U</td><td>10</td><td colspan="2">Analog Pin</td></tr>
<tr><td>FB_HV</td><td>13</td><td colspan="2">Analog Pin</td><td>Used to configure the high-voltage LDO when in custom mode (optional).
Must be left floating if not used.</td></tr>
<tr><td>SET_OVCH</td><td>22</td><td colspan="2" rowspan="3">Analog Pin</td><td rowspan="3">Used to configure the storage element protection thresholds when in custom mode (optional).
Must be connected to BUCK if not used.</td></tr>
<tr><td>SET_CHRDY</td><td>23</td></tr>
<tr><td>SET_OVDIS</td><td>24</td></tr>
<tr><td>ZMPP</td><td>25</td><td colspan="2">Analog Pin</td><td>Used to configure the ZMPPT (optional).
Must be left floating if not used.</td></tr>
<tr><td colspan="5">Control pins</td></tr>
<tr><td>ENHV</td><td>12</td><td>GND</td><td>BUCK</td><td>Enabling pin for the high-voltage LDO (See Table 8).</td></tr>
<tr><td>ENLV</td><td>18</td><td>GND</td><td>BUCK
to
BOOST</td><td>Enabling pin for the low-voltage LDO (See Table 8).
HIGH level is any voltage between BUCK and BOOST.</td></tr>
<tr><td colspan="5">Status pins</td></tr>
<tr><td>STATUS[2]</td><td>19</td><td>GND</td><td>BATT</td><td>Logic output. Asserted when the AEM performs a MPP evaluation.</td></tr>
<tr><td>STATUS[1]</td><td>20</td><td>GND</td><td>BATT</td><td>Logic output.
- HIGH:
  - during T_{CRIT} when in SHUTDOWN MODE.
  - as long as in PRIMARY BATTERY MODE.
- LOW otherwise.</td></tr>
<tr><td>STATUS[0]</td><td>21</td><td>GND</td><td>BATT</td><td>Logic output. Asserted when the LDOs can be enabled.</td></tr>
<tr><td colspan="5">Other pins</td></tr>
<tr><td>GND</td><td colspan="3">Exposed Pad</td><td>Ground connection, must be strongly tied to the PCB ground plane.</td></tr>
</table>

Table 2: Pins description (part 2)

3. Specifications

3.1. Absolute Maximum Ratings

Parameter		Min	Max	Unit
Operating junction temperature T_J		-40	85	°C
Storage temperature T_{stg}		-65	150	°C
Input voltage	SRC, BUFSRC, BATT, BAL, PRIM, BOOST, SWBOOST, HVOUT, ENLV, ZMPP	-0.3	5.50	V
	BUCK, SWBUCK, LVOUT, CFG[2:0], FB_PRIM_U, FB_PRIM_D, SELMPP[1:0], SET_OVDIS, SET_CHRDY, SET_OVCH, FB_HV, ENHV	-0.3	2.75	V

Table 3: Absolute maximum ratings

3.2. ESD Ratings

Parameter		Value	Unit
Electrostatic discharge V_{ESD}	Human-Body Model (HBM)[1]	± 1000	V
	Charged-Device Model (CDM)[2]	± 1000	V

Table 4: Absolute maximum ratings

1. ESD Human-Body Model (HBM) value tested according to JEDEC standard JS-001-2023.
2. ESD charged-Device Mode (CDM) value tested according to JEDEC standard JS-002-2022.

ESD CAUTION	
	ESD (ELECTROSTATIC DISCHARGE) SENSITIVE DEVICE These devices have limited built-in ESD protection and damage may thus occur on devices subjected to high-energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality

3.3. Thermal Resistance

Package	Parameter	Value	Unit
QFN 28-pin 5 x 5 mm	Junction-to-Ambient thermal resistance (θJA)	38.3	°C/W
	Junction-to-Case thermal resistance (θJC)	2.183	°C/W

Table 5: Thermal data

3.4. Electrical Characteristics at 25 °C

Symbol	Parameter	Condition	Min	Typ.	Max	Unit
Input voltage and input power						
$P_{SRC,CS}$	Minimum source power required to coldstart.			3		µW
$V_{SRC,CS}$	Minimum source voltage required to coldstart.			380		mV
V_{SRC}	Input voltage of the energy source (maximum given by the open-circuit voltage).	After cold start.	0.05[1]		5	V
I_{SRC}	Harvested current from the energy source.	L_{BOOST} = 10 µH.			110	mA
		L_{BOOST} = 22 µH.			50	mA
V_{MPP}	Target regulation voltage on SRC when extracting power.	After cold start.	0.05		4.5	V
DC-DC converters						
V_{BOOST}	Output voltage of the boost converter.	During normal operation.	2.2		4.5	V
V_{BUCK}	Output voltage of the buck converter.	During normal operation.	2	2.2	2.5	V
$V_{BUCK,RESET}$	Minimum voltage on BUCK before switching to DEEP SLEEP MODE (from any other mode).			1.8		V
I_{BUCK}	Total load current supplied by the BUCK converter (including LVOUT current I_{LV}).	L_{BUCK} = 10 µH.	0		20	mA
		L_{BUCK} = 4 µH.	0		50	mA
Storage element						
V_{BATT}	Voltage on the storage element.		0[2]		4.5	V
$V_{BAL,MIN}$	Minimum voltage required on the BAL pin for the balancing circuit to operate.			0.9		V
V_{PRIM}	Voltage on the primary battery.		0.6		V_{OVDIS} + 0.4 V	V
I_{PRIM}	Current from the primary battery.			20		mA
$V_{FB\_PRIM\_U}$	Feedback for defining the overdischarge voltage level on the primary battery.		0.15		1.1	V
V_{OVCH}	Maximum voltage accepted on the storage element before disabling the boost converter.	See Table 9.	2.30		4.50	V
V_{CHRDY}	Minimum voltage required on the storage element before enabling the LDO when coming from WAKE-UP MODE.	After cold start See Table 9.	2.250		4.45	V
V_{OVDIS}	Minimum voltage accepted on the storage element before switching to primary battery or entering SHUTDOWN MODE.	See Table 9.	2.20		4.40	V
I_Q	Quiescent current on BATT when the boost converter is not running.	V_{BATT} = 3 V LDOs disabled.		400		nA
		V_{BATT} = 3 V LDOs enabled.		600		nA
Low-Voltage LDO regulator						
V_{LV}[3]	Output voltage of the low-voltage LDO (see Table 9).		1.2		1.8	V
I_{LV}	Load current supplied by the low-voltage LDO.		0		20	mA

Table 6: Electrical characteristics (Part 1)

Symbol	Parameter	Condition	Min	Typ.	Max	Unit
High-Voltage LDO regulator						
V_{HV} [4]	Output voltage of the high-voltage LDO (see Table 9).		1.8		V_{OVDIS} - 0.3 V	V
I_{HV}	Load current supplied by the high-voltage LDO.		0		80	mA
Timing						
$T_{MPPT,WAIT}$	Time during which the AEM30940 stops pulling current on SRC before measuring the harvester open circuit voltage (V_{OC}).			5.12		ms
$T_{MPPT,MEASURE}$	Duration of V_{OC} measurement during MPP evaluations.			1.2		ms
$T_{MPPT,PERIOD}$	Period of the MPPT V_{OC} evaluations.			0.33		s
T_{CRIT}	Time before shutdown once STATUS[1] has been asserted (see Section 5.2.5 and Figure 13).		400	600	800	ms

Table 6: Electrical characteristics (Part 2)

1. *Minimum V_{SRC} value for harvesting capabilities after coldstart.*
2. *To stay in NORMAL MODE, V_{BATT} minimum voltage must stay above V_{OVDIS}.*
3. *The variability of V_{LV} at 1 mA is 1% (typical and preliminary result from simulations).*
4. *The variability of V_{HV} at 1 mA is 1.3% (typical and preliminary result from simulations).*

3.5. Recommended Operating Conditions

Symbol	Parameter		Min	Typ.	Max	Unit
External components						
C_{SRC}	BUFSRC pin decoupling capacitor.		8	10	22	µF
C_{BOOST}	Output capacitor of the boost converter.		10	22		µF
L_{BOOST}	Inductor of the boost converter.		4	22[1]		µH
C_{BUCK}	Output capacitor of the buck converter.		8	10		µF
L_{BUCK}	Inductor of the buck converter.		4	10	25	µH
C_{LV}	Low-voltage LDO regulator decoupling capacitor.		8	10	14	µF
C_{HV}	High-voltage LDO regulator decoupling capacitor.		8	10	14	µF
C_{BATT}	Capacitor connected on BATT if no storage element is connected (see Section 3.5.3).	LDOs disabled.	22			µF
		LDOs enabled.	150			µF
R_T	Optional - Sum of resistors for setting battery protection threshold voltages in custom mode. R_T = R1 + R2 + R3 + R4 (see Section 6.1).		1	10	100	MΩ
R_V	Optional - Sum of resistors for setting the output voltage of the high-voltage LDO in custom mode. R_V = R5 + R6 (see Section 6.1).		1	10	40	MΩ
R_{ZMPP}	Optional - Resistor for the ZMPPT configuration (see Section 6.4).		10		1M	Ω
R_P	Optional - Sum of resistors used to define the primary battery minimum voltage. R_P = R7 + R8 (see Section 6.3).		100		500	kΩ

Table 7: Recommended operating conditions

1.For optimal efficiency, a 22 µH inductor is recommend on LBOOST. However, a 10 µH inductor may also be used as an alternative when higher source current is required, though this will result in slightly lower efficiency.

3.5.1. External Inductors Information

The AEM30940 operates with two external inductors. All inductors must support a minimum switching frequency of 10 MHz. Using inductors with low equivalent series resistance (ESR) improves the power-conversion efficiency of both the boost and buck converters.

L_{BOOST}

The AEM30940 circuit is typically implemented with one of the following values on L_{BOOST}:

- 22 µH (peak current min. 115 mA) allows better efficiencies, especially at low SRC voltages.
- 10 µH (peak current min. 250 mA) allows higher current from SRC to BATT.

L_{BUCK}

With the recommended operating condition (10 µH inductor), the buck inductor L_{BUCK} must support a minimum peak current of 50 mA.

3.5.2. External Capacitors Information

The AEM30940 operates with 6 external miniature capacitors to ensure stable operation of the boost converter, buck converter, storage element output, and LDO outputs. Each capacitor serves as a local energy buffer that limits voltage fluctuations caused by switching activity or dynamic load transition.

To maintain optimal performances and minimized quiescent current, all capacitors must exhibit a low leakage current and follow the recommended nominal values listed in Table 7, with a tolerance of ± 20 %.

3.5.3. Storage Element Information

The energy storage element of the AEM30940 can be a rechargeable battery, a supercapacitor or a large capacitor. It should be selected so that its voltage never fall below V_{OVDIS} even during occasional peaks of the load current.

Connecting a decoupling capacitor (C_{BATT}) on the BATT pin is needed if:

- The internal resistance of the storage element cannot sustain V_{OVDIS} with the load current peaks;
- The application expects a disconnection of the storage element (e.g., because of a user removable connector, or during manufacturing), as the BATT pin cannot be left floating;
- The application does not require a storage element (see Section 6.5).

If the AEM30940 is used with no storage element or with a high-ESR storage element, the minimum C_{BATT} value required for optimal operation without voltage drops below V_{OVDIS} is:

- 150 µF if the LDOs are used.
- 22 µF is the LDOs are not used.

3.6. Typical Characteristics

3.6.1. Boost Conversion Efficiency for LBOOST = 22 µH

The following graph shows the boost converter efficiency from SRC to BATT, with L_{BOOST} = 22 µH (Coilcraft LPS4018-223MRB), including the loss of efficiency due to the AEM30940 quiescent current.

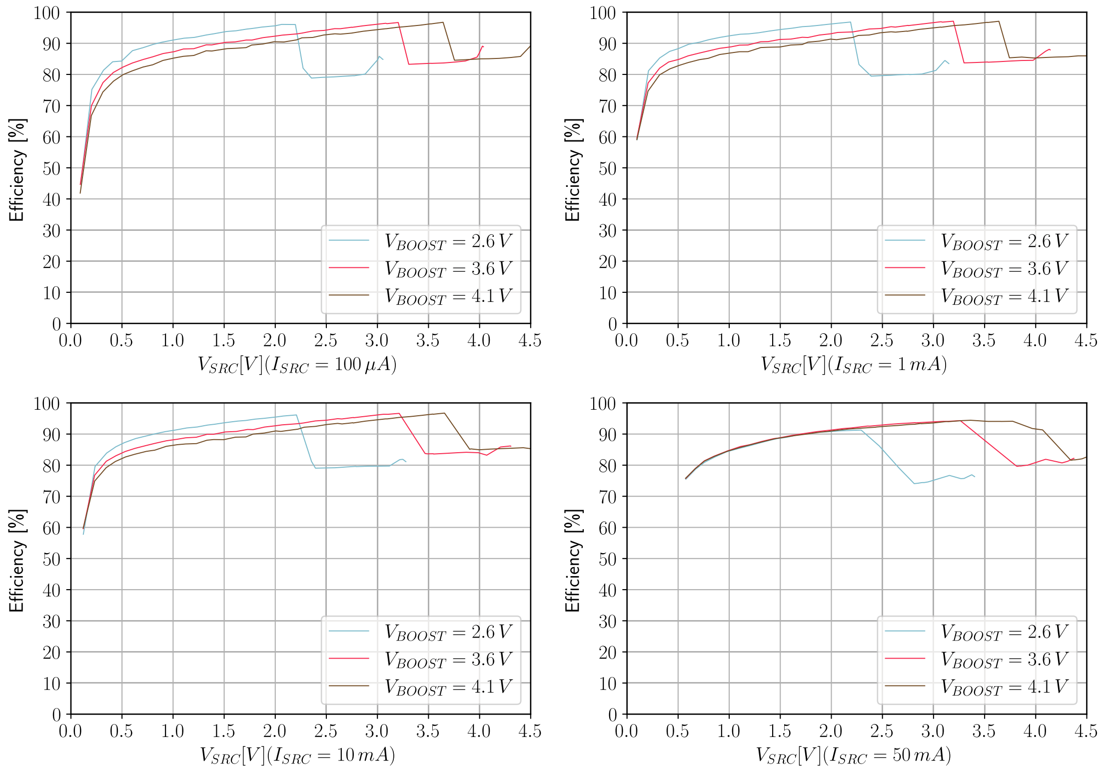


Figure 3: Boost converter efficiency with LBOOST = 22 µH (Coilcraft LPS4018-223MRB)

3.6.2. Boost Conversion Efficiency for LBOOST = 10 µH

The following graph shows the boost converter efficiency from SRC to BATT, with L_{BOOST} = 10 µH (Coilcraft LPS4018-103MRB), including the loss of efficiency due to the AEM30940 quiescent current.

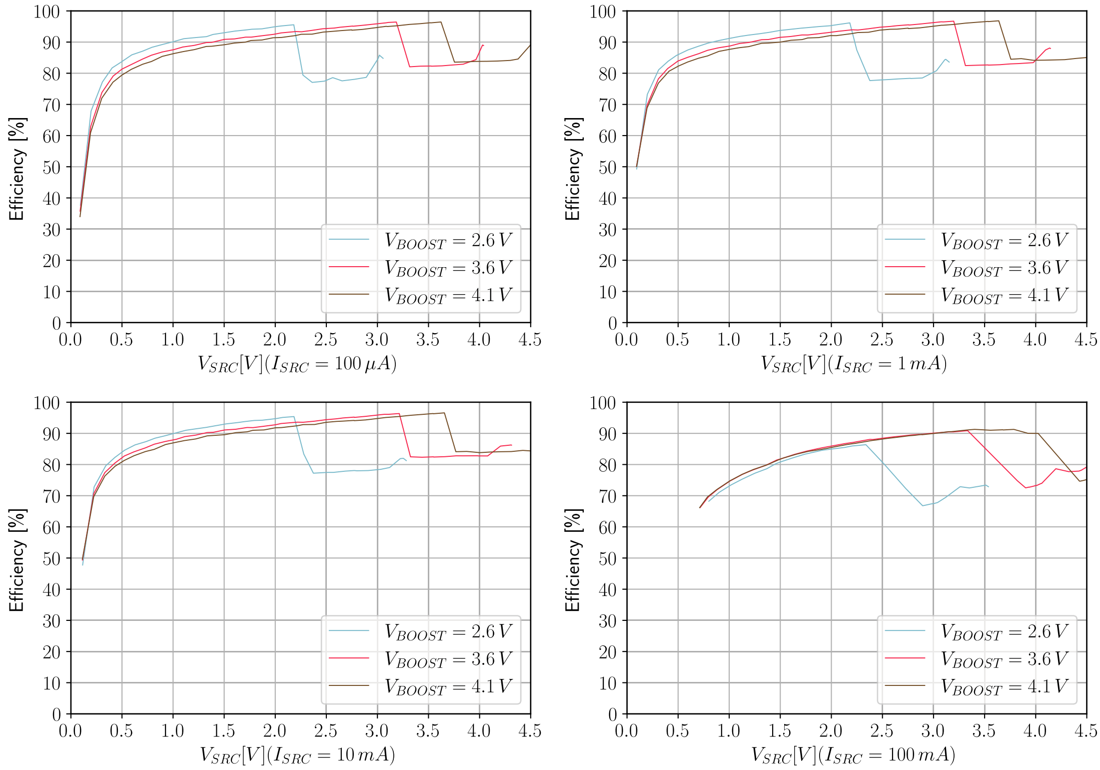


Figure 4: Boost converter efficiency with LBOOST = 10 µH (Coilcraft LPS4018-103MRB)

3.6.3. Buck Conversion Efficiency

The following graph shows the buck converter efficiency from BATT to BUCK, with L_{BUCK} = 10 µH (TDK MLZ1608N100LT000), with the AEM30940 quiescent current I_Q subtracted.

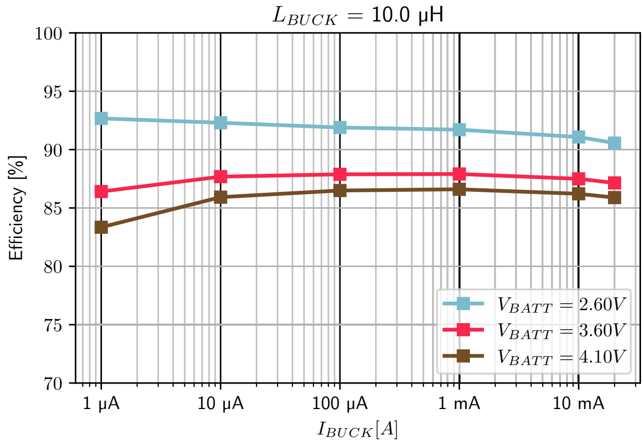


Figure 5: Buck converter efficiency with LBUCK = 10 µH (TDK MLZ1608N100LT000)

3.6.4. Quiescent Current

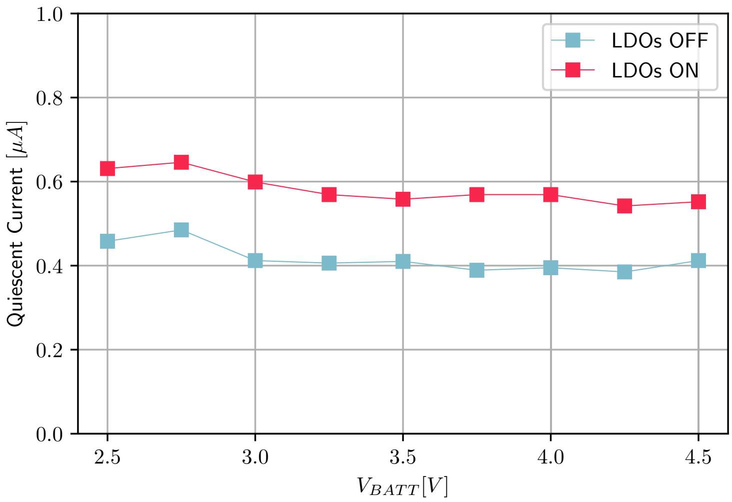


Figure 6: Quiescent current with LDOs on and off

3.6.5. High-voltage LDO Regulation

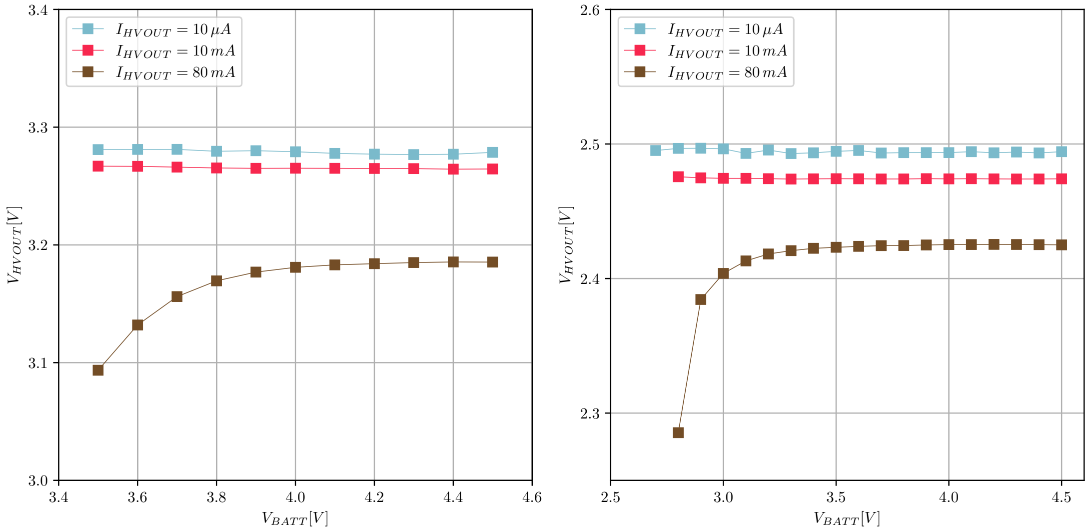


Figure 7: HVOUT at 3.3 V and 2.5 V

3.6.6. Low-voltage LDO Regulation

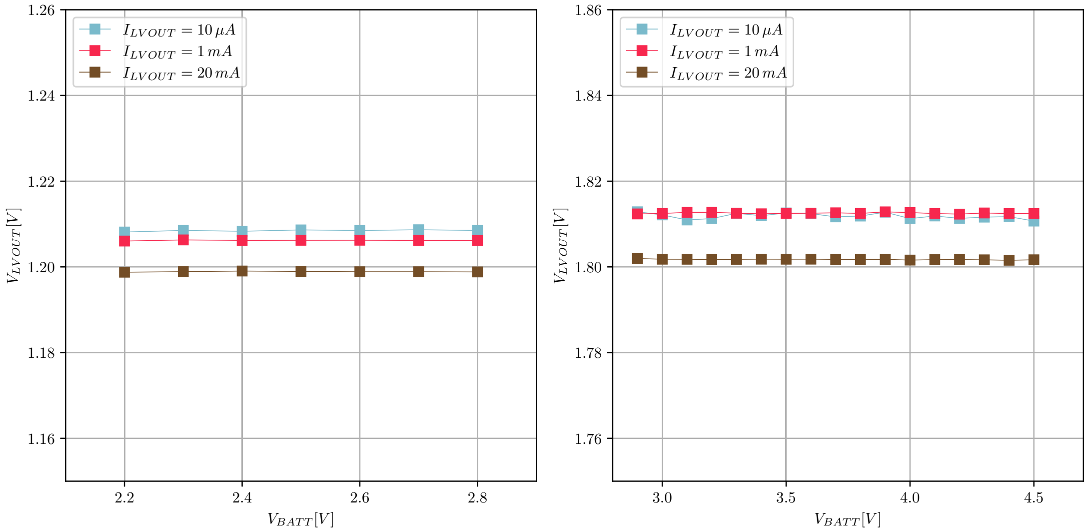


Figure 8: LVOUT at 1.2 V and 1.8 V

3.6.7. High-voltage LDO Efficiency

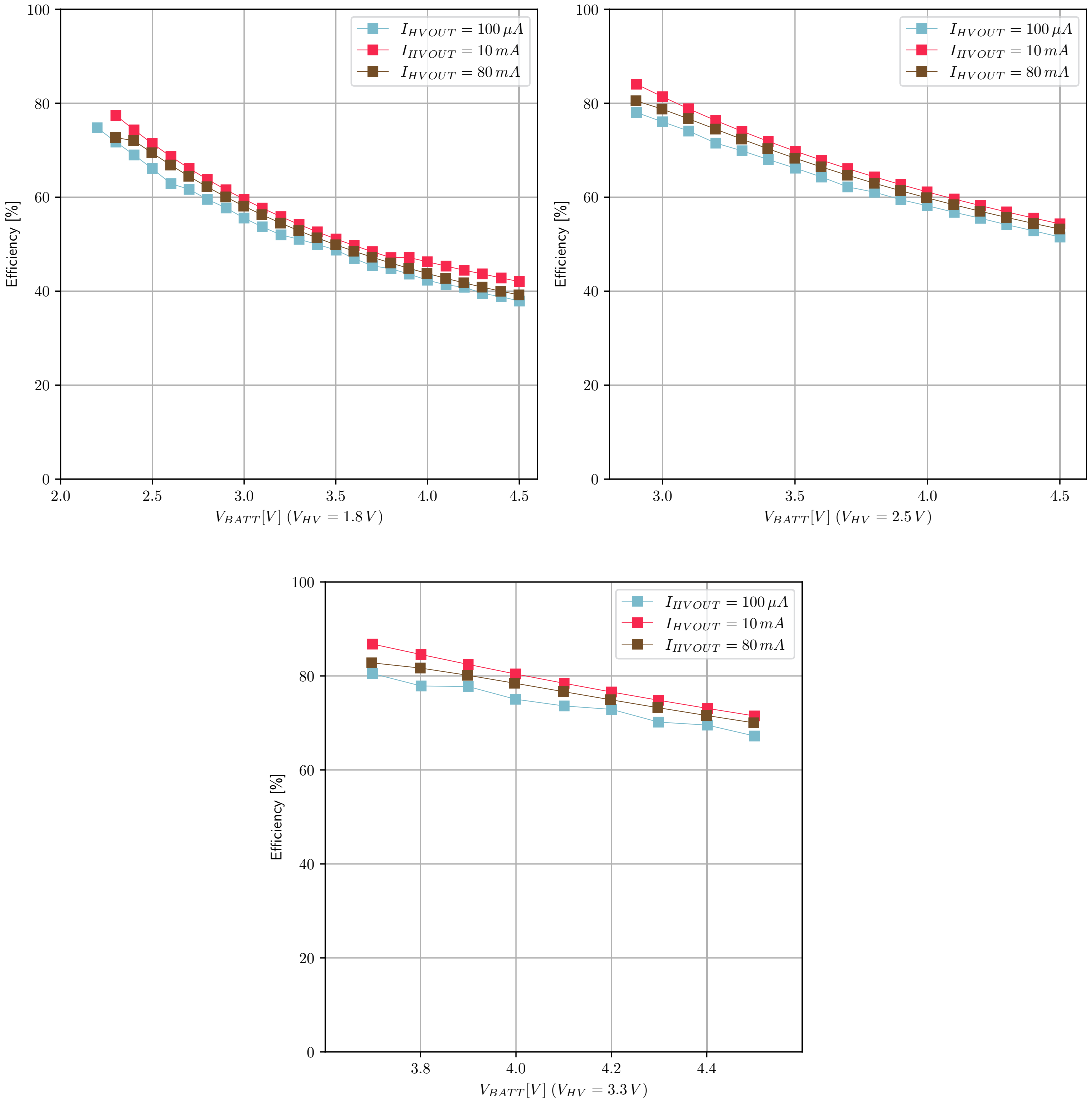


Figure 9: HVOUT efficiency at 1.8V, 2.5V and 3.3 V

The theoretical efficiency of an LDO can be calculated as Vout / Vin if quiescent current can be neglected with regards to the output current. For the high-voltage LDO, the theoretical efficiency is equal to V_{HV} / V_{BATT}.

3.6.8. Low-voltage LDO Efficiency

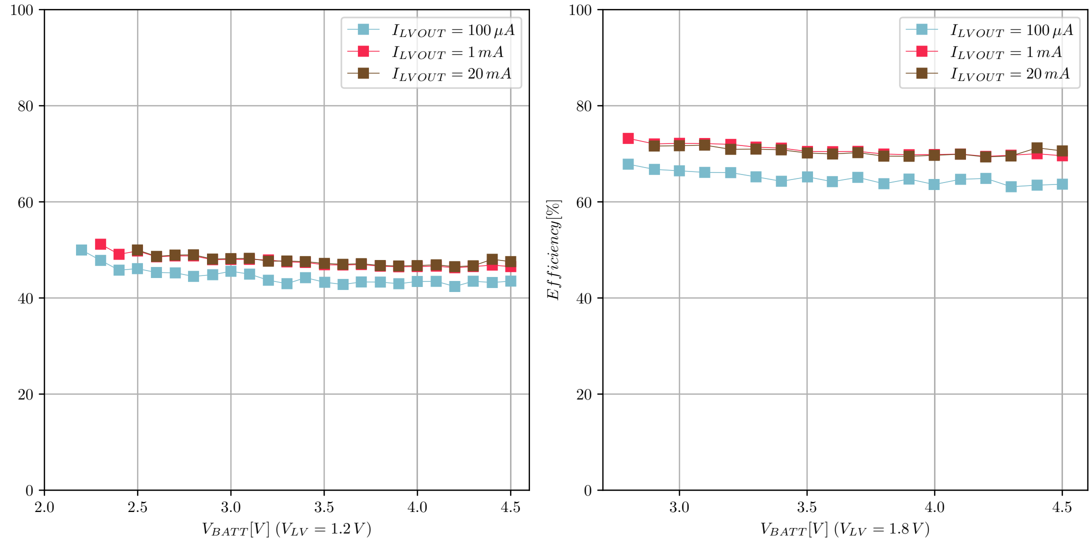


Figure 10: Efficiency of buck cascaded with LVOUT at 1.2 V and 1.8 V

The theoretical efficiency of an LDO can be calculated as V_{LV} / V_{BUCK}. Starting from the battery, the efficiency of the buck converter (ηbuck) has to be taken into account (see Figure 12).

The efficiency between V_{BATT} and V_{LV} is therefore equal to:

$$\eta_{BUCK} \cdot \frac{V_{LV}}{V_{BUCK}}$$

4. Functional Block Diagram

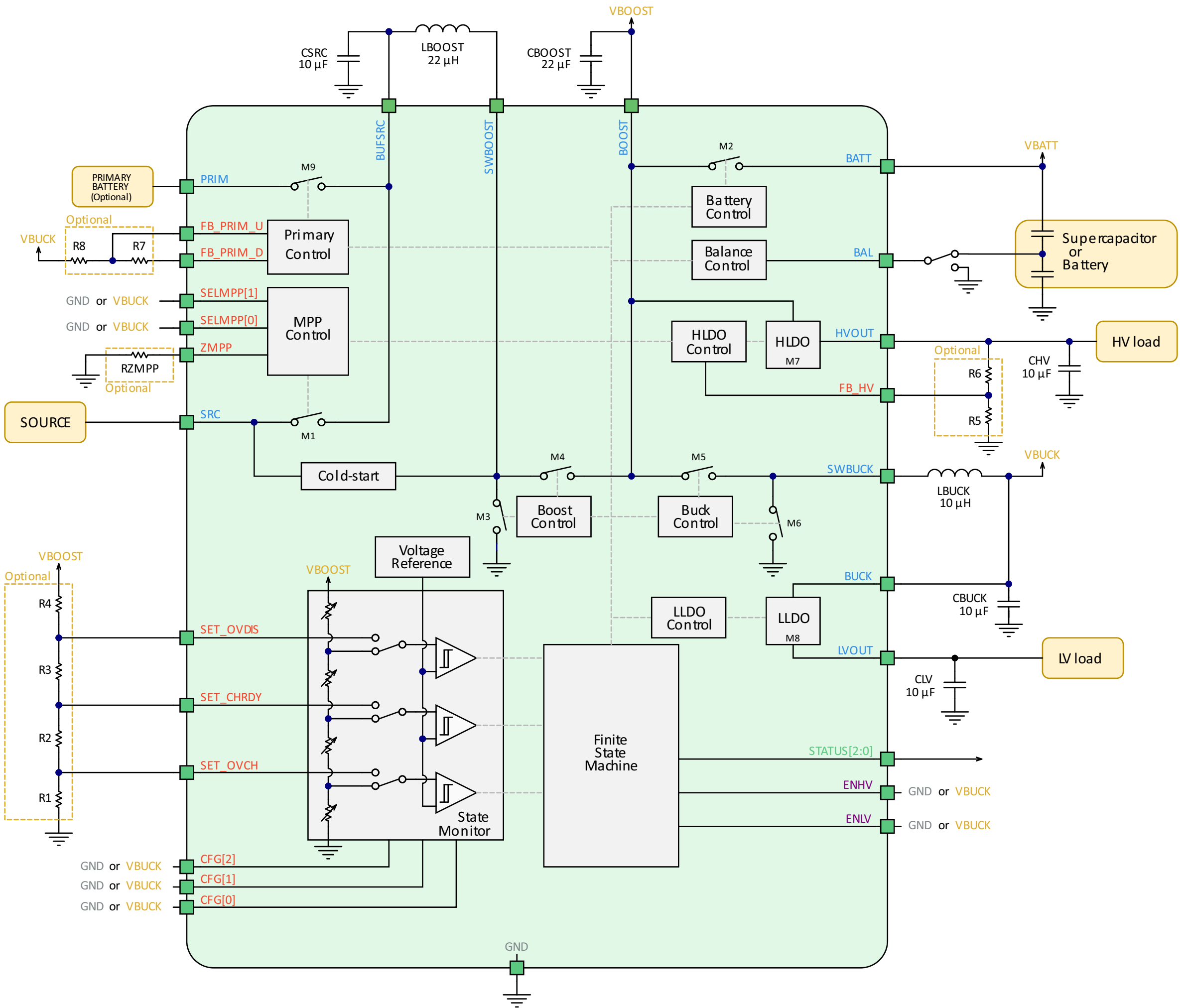


Figure 11: Functional block diagram

5. Theory of Operation

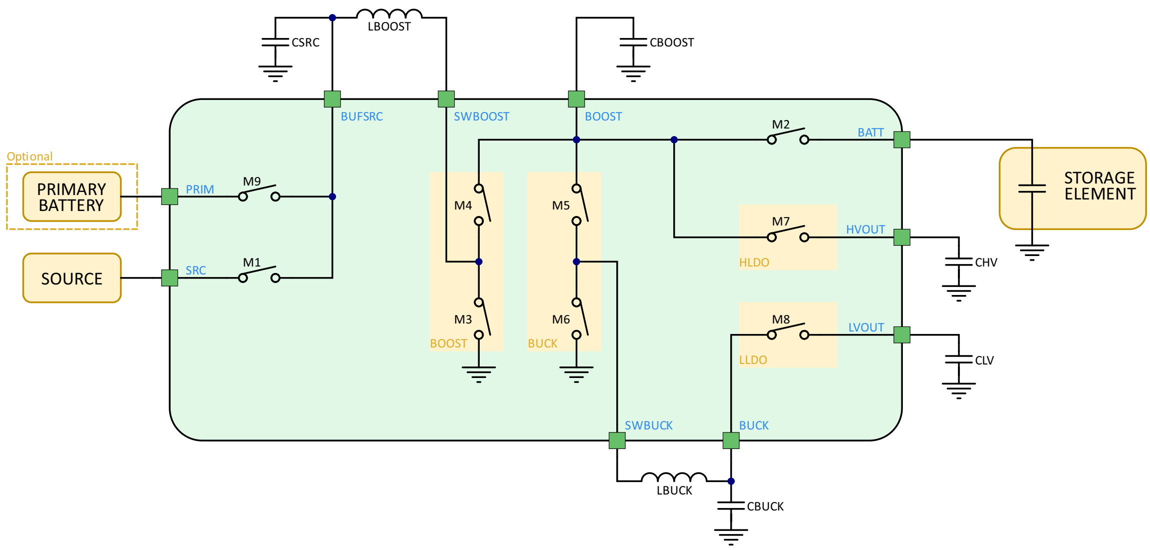


Figure 12: Simplified schematic view of the AEM30940

5.1. Power Converters

5.1.1. Boost Converter

The boost (or step-up) converter raises the voltage available at BUFSRC to a level suitable for charging the storage element, in the range of 2.2 V to 4.5 V, according to the system configuration. This voltage (V_{BOOST}) is available at the BOOST pin.

The switching transistors of the boost converter are M3 and M4, with the switching node available externally at SWBOOST. The reactive power components of this converter are the external inductor L_{BOOST} and the external capacitor C_{BOOST}.

The MPPT control circuit (see Section 5.4) periodically disconnects the source on SRC pin from the BUFSRC pin with the transistor M1 during $T_{MPPT,WAIT}$ in order to let the SRC voltage rise to the open-circuit voltage (V_{OC}). The AEM30940 then measures the open-circuit voltage of the harvester on SRC during $T_{MPPT,MEASURE}$ and define the new optimal SRC regulation voltage. This MPP evaluation is repeated every $T_{MPPT,PERIOD}$.

BUFSRC is decoupled by the capacitor C_{SRC}, which smooths the voltage against the current pulses induced by the boost converter.

The storage element is connected to the BATT pin. Its voltage is named V_{BATT}. This node is linked to BOOST through the transistor M2. In NORMAL MODE (see Section 5.2.2), this transistor effectively shorts the battery to the BOOST node ($V_{BATT} = V_{BOOST}$). When energy harvesting is occurring, the boost converter delivers a current that is shared between the battery and the LDOs. M2 is opened to disconnect the storage element when V_{BATT} reaches V_{OVDIS}. However, in such a scenario, the AEM30940 offers the possibility of connecting a primary battery to recharge V_{BATT} up to V_{CHRDY}. In this case, the transistor M9 connects PRIM to BUFSRC and the transistor M1 is opened to disconnect the SRC input pin as explained in the PRIMARY BATTERY MODE section.

To ensure the boost converter operates correctly:

- when using PRIM as input of the boost converter (in PRIMARY BATTERY MODE), the voltage of the primary battery must remain below V_{OVDIS} + 0.4 V (see Table 6).
- when using SRC as input of the boost converter, the harvester maximum current must not exceed I_{SRC} maximum value (see Table 6).

More explanations about the different modes can be found in Section 5.2.

5.1.2. Buck Converter

The buck (or step-down) converter lowers the voltage from V_{BOOST} to a constant V_{BUCK} value of 2.2 V. This voltage is available at the BUCK pin. The switching transistors of the buck converter are M5 and M6, with the switching node available externally at SWBUCK. The reactive power components of the buck converter are the external inductor L_{BUCK} and the external capacitor C_{BUCK}.

5.1.3. LDO Outputs

Two Low Drop-Out linear regulators are available to supply loads at different operating voltages:

- Through M7, BOOST supplies the high-voltage LDO that powers its load through HVOUT. This regulator delivers a clean voltage named V_{HV}. When using the built-in configuration modes, an output voltage of 1.8 V, 2.5 V or 3.3 V can be configured through CFG[2:0]. When using the custom configuration mode, V_{HV} is adjustable between 1.8 V and V_{OVDIS} - 0.3 V. The output is decoupled by the external capacitor C_{HV}.
- Through M8, V_{BUCK} supplies the low-voltage LDO that powers its load through LVOUT. This regulator delivers a clean voltage named V_{LV} of 1.2 V or 1.8 V configured through CFG[2:0]. The output is decoupled by the external capacitor C_{LV}.

See Table 6 for HVOUT and LVOUT maximum current values (respectively I_{HV} and I_{LV}).

Both the high-voltage and the low-voltage outputs can be dynamically enabled or disabled respectively with the logic control pins ENHV and ENLV (see table below).

ENLV	LVOUT
L	Disabled
H	Enabled

ENHV	HVOUT
L	Disabled
H	Enabled

Table 8: LDOs configurations

5.2. Operating Modes

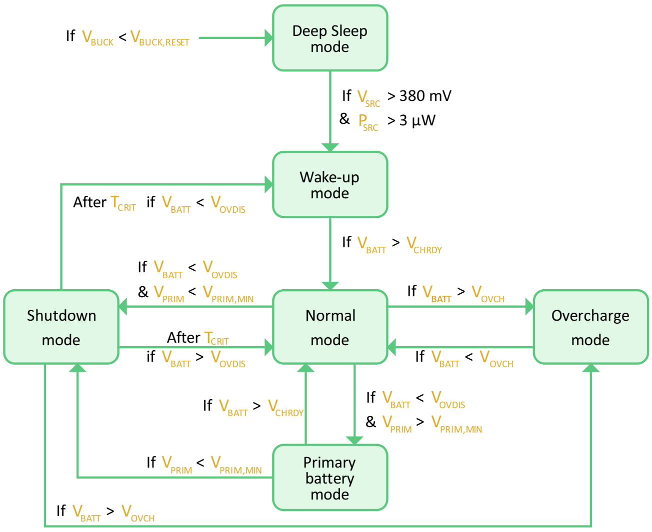


Figure 13: Diagram of the AEM30940 modes

5.2.1. Deep Sleep & Wake Up Modes

The DEEP SLEEP MODE is a state where all nodes are deeply discharged and there is no available energy to be harvested.

The AEM30940 enters DEEP SLEEP MODE if V_{BUCK} falls below $V_{BUCK,RESET}$ (see Table 6).

In this mode, as soon as the required cold-start voltage of 380 mV and the required power of 3 µW becomes available on SRC, the WAKE-UP MODE is activated. For RF sources, the input power at the antenna is defined by the matching network and the associated rectifier (see Section 5.3). V_{BOOST} and V_{BUCK} rise up to a voltage of 2.2 V. V_{BOOST} then rises up to V_{OVCH}.

At this stage, both LDOs are internally disabled. Therefore, STATUS[0] is low as shown in Figure 19 and Figure 20.

When V_{BOOST} reaches V_{OVCH}, two scenarios are possible:

- In the first scenario, a supercapacitor or a capacitor having a voltage lower than V_{CHRDY} is connected to the BATT node (see Section 5.2.1.1).
- In the second scenario, a charged battery is connected to the BATT node (see Section 5.2.1.2).

5.2.1.1. Supercapacitor as a Storage Element

If the storage element is a supercapacitor, the storage element may need to be charged from 0 V. The boost converter charges BATT from the input source and by modulating the conductance of M1 and M2. During the charge of the BATT node, both LDOs are disabled and STATUS[0] is low. When V_{BATT} reaches V_{CHRDY}, the circuit enters NORMAL MODE, STATUS[0] is asserted and the LDOs can be enabled by the user using ENLV and ENHV control pins as shown in Figure 19.

5.2.1.2. Battery as a Storage Element

If the storage element is a battery but its voltage is lower than V_{CHRDY}, the storage element first needs to be charged until it reaches V_{CHRDY}. This allows a safety margin to ensure that the storage element is able to provide the required power before enabling the outputs (LDOs).

Once V_{BATT} exceeds V_{CHRDY}, or if the battery was initially charged above V_{CHRDY}, the circuit enters NORMAL MODE. STATUS[0] is asserted and the LDOs can be dynamically enabled or disabled through ENLV and ENHV as shown in Figure 20.

5.2.2. Normal Mode

Once the AEM enters NORMAL MODE, it stays in this mode as long as the following condition is met:

$$V_{OVDIS} < V_{BATT} < V_{OVCH}$$

The AEM30940 will switch to another mode in the following cases:

- V_{BATT} increases above V_{OVCH} because the source provides more power than the load consumes. The circuit enters OVERCHARGE MODE, as explained in Section 5.2.3.
- V_{BATT} falls below V_{OVDIS} due to a lack of power from the source. In this case, either the circuit enters SHUTDOWN MODE as explained in Section 5.2.5, or, if a charged primary battery is connected on PRIM, the circuit enters PRIMARY BATTERY MODE as explained in Section 5.2.4.

5.2.3. Overcharge Mode

When V_{BATT} reaches V_{OVCH}, the battery charge is complete. The AEM maintains V_{BATT} around V_{OVCH}, with an hysteresis of a few mV as shown in Figure 21, to prevent damage to the storage element and to the internal circuitry. In this configuration, the boost converter is periodically activated to maintain V_{BATT} and the LDO output voltages are available. Moreover, when the boost converter is not activated, the transistor M1 in Figure 12 is opened to prevent current from the source to the storage element when V_{SRC} is higher than V_{OVCH}.

5.2.4. Primary Battery Mode

When V_{BATT} drops below V_{OVDIS}, the circuit compares the voltage on PRIM with the voltage on FB_PRIM_U to determine whether a charged primary battery is connected on PRIM. The voltage on FB_PRIM_U is set thanks to two optional resistors as explained in Section 6.3.

If the following formula is true, the circuit considers the primary battery as available and the circuit enters PRIMARY BATTERY MODE.

$$\frac{V_{PRIM}}{4} > V_{FB\_PRIM\_U}$$

In that mode, transistor M1 is opened and the primary battery is connected to BUFSRC through transistor M9 to become the source of energy of the AEM30940. STATUS[1] is asserted as long as the chip is in PRIMARY BATTERY MODE.

The AEM remains in this mode until:

- V_{BATT} reaches V_{CHRDY}. At that point, the circuit enters NORMAL MODE.
- The primary battery is depleted (above formula becomes false). The circuit switches to SHUTDOWN MODE.

If no primary battery is used in the application, PRIM, FB_PRIM_U and FB_PRIM_D must be tied to GND.

5.2.5. Shutdown Mode

When V_{BATT} drops below V_{OVDIS} and no power is available from a primary battery, the circuit enters SHUTDOWN MODE, as shown in Figure 22, to prevent deep discharge that could damage the storage element and make the LDOs unstable. The circuit asserts STATUS[1] to warn the application that a shutdown may occur. Both LDO regulators remain enabled during T_{CRIT} (about 600 ms).

If no primary battery is used, this mechanism allows the application circuit, whether it is powered on LVOUT or HVOUT, to trigger an interrupt by the low to-high transition of STATUS[1], and to take all appropriate actions before LVOUT and HVOUT are disabled.

After T_{CRIT}:

- If V_{BATT} has recovered above V_{OVDIS}, the AEM switches back to NORMAL MODE.
- If V_{BATT} has not recovered above V_{OVDIS}, the circuit enters WAKE-UP MODE. Both LDOs are disabled and BATT is disconnected from BOOST to avoid damaging the battery due to the over-discharge. From now on, the AEM must go through the wake-up procedure described in the Section 5.2.1.

If V_{BATT} reaches V_{OVCH} while in SHUTDOWN MODE, the AEM30940 switches directly to OVERCHARGE MODE without waiting the end of T_{CRIT}.

5.3. Matching Network and Rectifier

To connect AC sources to the AEM30940, an external rectifier is required as well as a matching network in the case of RF energy harvesting.

The goal of the matching network is to modify the impedance relationship between the RF source and the rectifier in order to optimize the power transfer over a frequency band and an input power range.

For RF applications, external high-frequency rectifiers as well as matching networks are available for the following RF bands:

- 868 MHz
- 915 MHz
- 2.4 GHz

Please contact e-peas for more information about matching network and rectifier designs and results.

5.4. Maximum Power Point Tracking

During NORMAL MODE, SHUTDOWN MODE and a part of WAKE-UP MODE, the boost converter is regulated thanks to an internal MPPT (Maximum Power Point Tracking) module.

The SRC regulation voltage (V_{MPP}) is defined during the MPPT evaluations as a constant fraction of the source open-circuit voltage (V_{OC}). The ratio between V_{MPP} and V_{OC} can be configured with the SELMPP[1:0] pins.

During the MPPT evaluations, the AEM30940 stops pulling current from the source (SRC) during $T_{MPPT,WAIT}$ (5.12 ms), in order to let the source rise to its open-circuit voltage (V_{OC}), which is then measured during $T_{MPPT,MEASURE}$ (1.2 ms). The new source target regulation voltage (V_{MPP}) is then defined as the configured fraction (MPPT ratio) of the measured V_{OC}. The evaluation is repeated every $T_{MPPT,PERIOD}$ (0.33 s). This way, the MPPT module adapts to the harvester variations due to varying ambient conditions. The behavior of the MPPT module is shown in Figure 14.

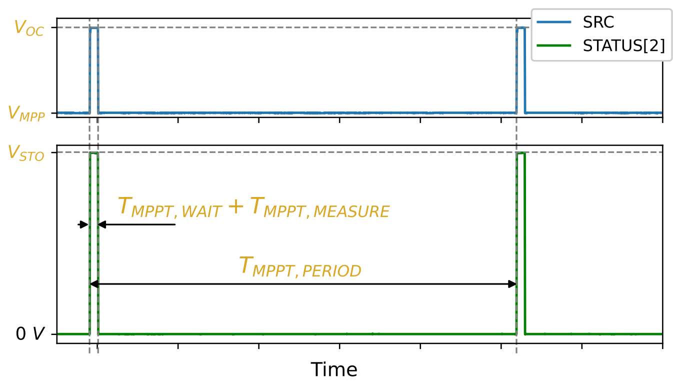


Figure 14: MPP evaluation behavior

With the exception of this MPP evaluation process, the source voltage V_{SRC} is continuously compared to V_{MPP}:

- When V_{SRC} exceeds V_{MPP} by a small hysteresis, the boost converter is switched on, extracting electric charges from the source, thus lowering its voltage.
- When V_{SRC} falls below V_{MPP} by a small hysteresis, the boost converter is switched off, allowing the harvester to accumulate new electric charges into C_{SRC}, which voltage rises.

This way, the boost converter regulates its input voltage so that the electric current (or flow of electric charges) that enters the boost converter yields the best power transfer from the harvester under any ambient conditions. The AEM30940 supports any V_{MPP} level in the range from 50 mV to 4.5 V. It offers a choice of three values for the V_{MPP} / V_{OC} ratio through the configuration pins SELMPP[1:0] as shown in Table 10. It is also possible to regulate the source voltage by matching the input impedance of the BOOST converter with an impedance connected to the ZMPP terminal thanks to the ZMPPT feature, by setting SELMPP[1:0] to HH (see Section 6.4). The status of the MPPT controller is reported through one dedicated status pins (STATUS[2]). This status pin is asserted when a MPPT module periodic V_{OC} evaluation is being performed.

5.5. Storage Element Balancing Circuit for Dual-cell Supercapacitor

When using a dual-cell supercapacitor (with the cells in series) it is necessary to keep both cells at similar voltages to avoid damage due to a potential over-voltage on one cell. This is ensured by the AEM30940 storage element balancing circuit (BAL pin).

If a battery, a capacitor or a single-cell supercapacitor is connected on BATT, the balancing circuit is not needed. BAL must be connected to GND to disable the storage element balancing circuit.

If a dual-cell supercapacitor is connected on BATT, BAL must be connected to the node between the two cells of the supercapacitor. The storage element balancing circuit compensates for any mismatch of the two cells that could overcharge one of the two cells. It ensures that BAL voltage remains close to V_{BATT} / 2.

The balancing circuit works as follows:

- The balancing circuit is enabled only when $V_{BAL} > V_{BAL,MIN}$ (see Table 6).
- If $V_{BAL} > \frac{V_{BATT}}{2}$: the AEM30940 enables a switch between BAL and GND to momentarily discharge the bottom cell to GND until balance is restored.
- If $V_{BAL} < \frac{V_{BATT}}{2}$: the AEM30940 enables a switch between BATT and BAL to transfer charge from the top cell to the bottom cell until balance is restored.

6. System Configuration

6.1. Battery and LDOs Configuration

Configuration pins			Storage element threshold voltages			LDOs output voltages		Typical use
CFG [2]	CFG [1]	CFG [0]	V_{OVCH}	V_{CHRDY}	V_{OVDIS}	V_{HV}	V_{LV}	
H	H	H	4.12 V	3.67 V	3.60 V	3.3 V	1.8 V	Li-ion battery
H	H	L	4.12 V	4.04 V	3.60 V	3.3 V	1.8 V	Solid state battery
H	L	H	4.12 V	3.67 V	3.01 V	2.5 V	1.8 V	Li-ion/NiMH battery
H	L	L	2.70 V	2.30 V	2.20 V	1.8 V	1.2 V	Single-cell supercapacitor
L	H	H	4.50 V	3.67 V	2.80 V	2.5 V	1.8 V	Dual-cell supercapacitor
L	H	L	4.50 V	3.92 V	3.60 V	3.3 V	1.8 V	Dual-cell supercapacitor
L	L	H	3.63 V	3.10 V	2.80 V	2.5 V	1.8 V	LiFePO4 battery
L	L	L	Custom mode - Programmable with R1 to R6.				1.8 V	

Table 9: Usage of CFG[2:0]

DISCLAIMER: storage element thresholds provided in the table above are indicative to support a wide range of storage element variants. They are provided as is to the best knowledge of e-peas's application laboratory. They should not replace the actual values provided in the storage element manufacturer's specifications and datasheet.

Through three configuration pins (CFG[2:0]), the user can set a particular operating mode from a range that covers most application requirements, as shown in Table 9, without any dedicated external component. The three threshold levels are defined as:

- V_{OVCH}: maximum voltage accepted on the storage element before disabling the boost converter.
- V_{CHRDY}: minimum voltage required on the storage element after a cold start before enabling the LDOs.
- V_{OVDIS}: minimum voltage accepted on the storage element before considering the storage element as depleted.

See Section 5 for more information about the purposes of these thresholds.

The two LDOs output voltages are called V_{HV} and V_{LV} for the high and low output voltages respectively. Seven combinations of these voltage levels are hard-wired and selectable through the CFG[2:0] configuration pins, covering most application cases. For other V_{OVCH}, V_{CHRDY}, V_{OVDIS} and V_{HV} voltages combinations, a custom mode is available. In this mode, the user can define those voltages with resistors, connected to the pins named SET_OVDIS, SET_CHRDY, SET_OVCH and FB_HV.

When the custom mode is not used, SET_OVDIS, SET_CHRDY and SET_OVCH pins must be connected to V_{BUCK} and FB_HV must be left floating.

6.1.1. Custom Mode

When CFG[2:0] are tied to GND, the custom mode is selected. All six resistors, shown in Figure 15, are used to configure the custom mode as follows:

V_{OVCH}, V_{CHRDY} and V_{OVDIS} are defined thanks to R1, R2, R3 and R4. The resistors are calculated as follows:

- $R_T = R1 + R2 + R3 + R4$
- $1M\Omega \leq R_T \leq 100M\Omega$
- $R1 = R_T \cdot \frac{1V}{V_{OVCH}}$
- $R2 = R_T \cdot \left(\frac{1V}{V_{CHRDY}} - \frac{1V}{V_{OVCH}}\right)$
- $R3 = R_T \cdot \left(\frac{1V}{V_{OVDIS}} - \frac{1V}{V_{CHRDY}}\right)$
- $R4 = R_T \cdot \left(1 - \frac{1V}{V_{OVDIS}}\right)$

V_{HV} is defined thanks to R5 and R6. The resistive divider is configured as follows:

- $R_V = R5 + R6$
- $1M\Omega \leq R_V \leq 40M\Omega$
- $R5 = R_V \cdot \frac{1V}{V_{HV}}$
- $R6 = R_V \cdot \left(1 - \frac{1V}{V_{HV}}\right)$

NOTE: If ENHV = L (HVOUT is disabled), R5 and R6 are not needed, FB_HV should be left floating.

The resistors should have high values to make the current flowing through them negligible. Moreover, the following constraints must be met to ensure the functionality of the chip:

- $V_{CHRDY} + 0.05V \leq V_{OVCH} \leq 4.5V$
- $V_{OVDIS} + 0.05V \leq V_{CHRDY} \leq V_{OVCH} - 0.05V$
- $2.2V \leq V_{OVDIS}$
- $V_{HV} \leq V_{OVDIS} - 0.3V$

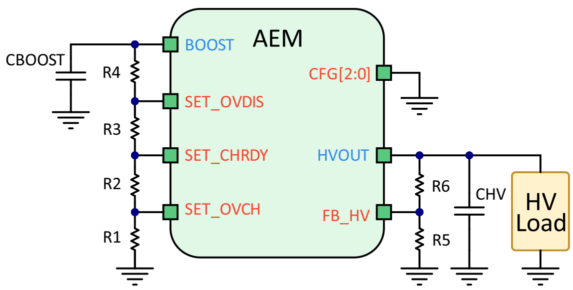


Figure 15: Custom configuration resistors

6.2. MPPT Configuration

Two dedicated configuration pins, SELMPP[1:0], allow selecting the MPP tracking ratio based on the characteristic of the input power source.

Configuration pins		MPPT ratio
SELMPP[1]	SELMPP[0]	V_{MPP} / V_{OC}
L	L	50%
L	H	65%
H	L	80%
H	H	ZMPPT

Table 10: Usage of SELMPP[1:0]

6.3. Primary Battery Configuration

To use the primary battery, it is mandatory to determine $V_{PRIM,MIN}$, the voltage at which the primary battery is considered as fully depleted. The circuit uses a resistive divider between BUCK and FB_PRIM_D to define the voltage on FB_PRIM_U as $V_{PRIM,MIN}$ divided by 4. During $V_{PRIM,MIN}$ evaluation, the circuit connects FB_PRIM_D to GND. When $V_{PRIM,MIN}$ is not evaluated, FB_PRIM_D is left floating to avoid quiescent current on the resistive divider. The resistors are calculated as follows:

- R_P = R7 + R8
- $100\mathrm{k}\Omega \leq R_P \leq 500\mathrm{k}\Omega$
- $R7 = \frac{V_{PRIM\_MIN}}{4} \cdot R_P \cdot \frac{1}{2.2V}$
- $R8 = R_P - R7$

NOTE: FB_PRIM_U, FB_PRIM_D and PRIM must be tied to GND if no primary battery is used.

6.4. ZMPPT Configuration

Instead of working at a ratio of the open-circuit voltage, the AEM30940 can regulate the input impedance of the BOOST converter so that it matches a constant impedance (R_{ZMPP}) connected between the ZMPP pin and GND. In this case, the AEM30940 regulates V_{SRC} at a voltage equal to the product of the ZMPP impedance and the current available at the SRC input.

- $10\Omega \leq R_{ZMPP} \leq 1\mathrm{M}\Omega$

6.5. No-battery Configuration

If the application does not use a storage element, the PCB must include a capacitor on the BATT pin. See Section 3.5.3 for C_{BATT} value.

The storage element may not be necessary in the following cases:

- If the harvested energy source is permanently available and covers the application purposes.
- If the application does not need to store energy when the harvested energy source is not available.

6.6. Supplying an Application Circuit with BUCK

It is possible to supply an application circuit directly from BUCK, with the benefit of high BATT to BUCK efficiency, provided that the following conditions are met:

- The application circuit can be supplied from a voltage in the 2.0 V - 2.5 V range (V_{BUCK} is typically 2.2 V with ripple, see Table 6).
- The sum of the following currents must be below the maximum I_{BUCK} value (see Table 6):
  - Current of the load connected to BUCK.
  - Current of the load connected to LVOUT.
- The application circuit on BUCK does not pull current during the AEM30940 cold start.

To satisfy the last condition, the following circuit may be implemented:

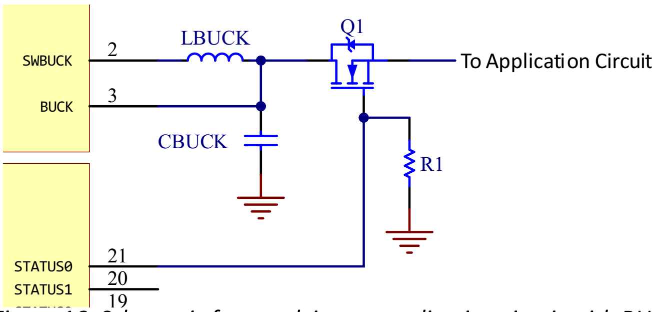


Figure 16: Schematic for supplying an application circuit with BUCK

Q1 is a N-MOSFET, whose gate is driven by STATUS[0] with R1 as a pull-down resistor. When the AEM30940 is in DEEP SLEEP MODE or in WAKE-UP MODE, STATUS[0] is LOW (see Section 5.2), ensuring that Q1 is non-conducting, and thus that the application circuit is not supplied.

When the AEM30940 switches from WAKE-UP MODE to NORMAL MODE, STATUS[0] is HIGH, making Q1 conducting. The application circuit is then supplied by BUCK, and remains so when the AEM30940 is in NORMAL MODE, OVERCHARGE MODE, PRIMARY BATTERY MODE and SHUTDOWN MODE.

Q1 must be chosen as follows:

- Low Gate-Source Leakage I_{GSS}.
- Low Zero Gate Voltage Drain Current I_{DSS}.
- Drain-Source On-State Resistance $R_{DS(on)}$ low enough to supply application circuit with an acceptable voltage drop.
- V_{GS} maximum voltage must be above V_{OVCH} (STATUS[0] HIGH voltage is V_{BOOST}).
- Maximum gate-source threshold voltage $V_{GS(th),MAX}$ matches the following, with $V_{BUCK,MAX}$ being V_{BUCK} maximum value stated in Table 6:

$$V_{GS(th),MAX} < V_{OVDIS} - V_{BUCK,MAX}$$

7. Typical Application Circuits

7.1. Example Circuit 1

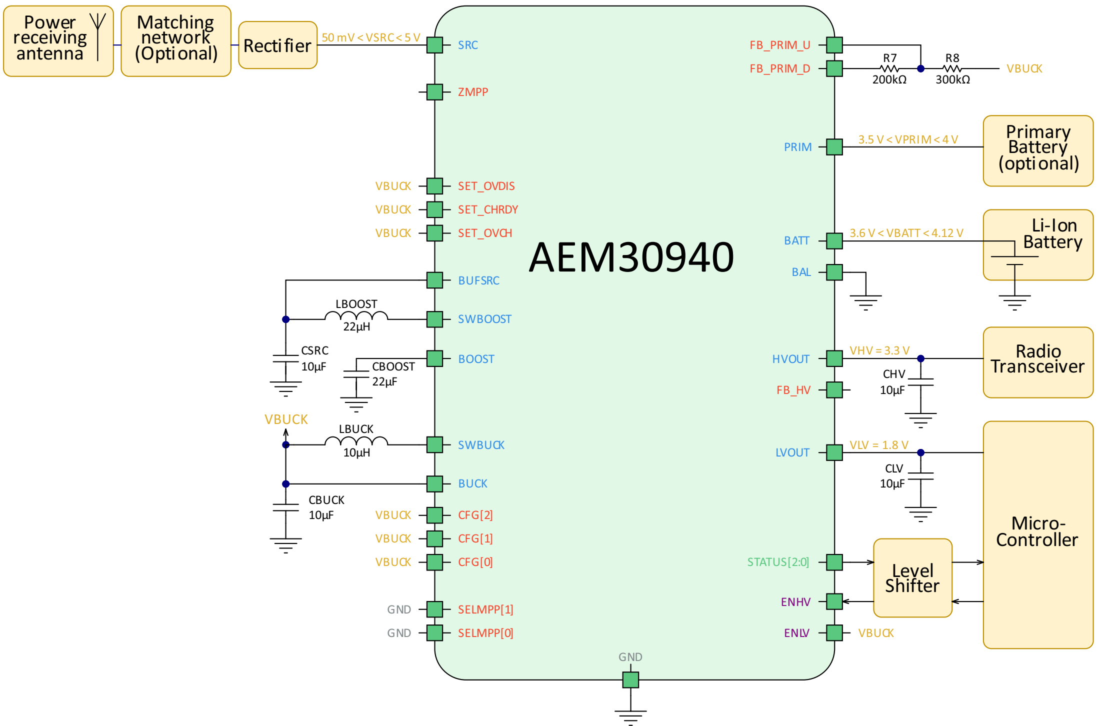


Figure 17: Typical application circuit 1

The energy source is a RF source and the storage element is a standard Li-ion battery cell. The radio communication is supplied by HVOUT set at 3.3 V. The micro-controller that controls the application is supplied by LVOUT set at 1.8 V.

This circuit uses a pre-defined AEM configuration, typical of systems that use standard components for radio and energy storage.

The operating mode pins are set as follows:

- CFG[2:0] = HHH (all to V_{BUCK})

Referring to Table 9, in this mode, the threshold voltages are:

- V_{OVCH} = 4.12 V
- V_{CHRDY} = 3.67 V
- V_{OVDIS} = 3.60 V

Moreover, the LDOs output voltages are:

- V_{HV} = 3.3 V
- V_{LV} = 1.8 V

A primary battery is also connected as a back-up solution. The minimal level allowed on this battery is set at 3.5 V. Following equations from Section 6.3:

- $R_P = 0.5\text{M}\Omega$
- $R7 = \frac{3.5\text{V}}{4} \cdot 0.5\text{M}\Omega \cdot \frac{1}{2.2\text{V}} = 200\text{k}\Omega$
- $R8 = 0.5\text{M}\Omega - 200\text{k}\Omega = 300\text{k}\Omega$

The MPPT configuration pins SELMPP[1:0] are tied to GND (logic LOW), thus, selecting an MPPT ratio of 50%.

The LVOUT LDO output is enabled by tying ENLV to V_{BUCK}. The micro-controller is supplied by LVOUT, that is enabled when V_{BATT} and V_{BOOST} voltage rise above V_{CHRDY}.

The application software can enable or disable the radio transceiver supply with a GPIO connected to ENHV.

7.2. Example Circuit 2

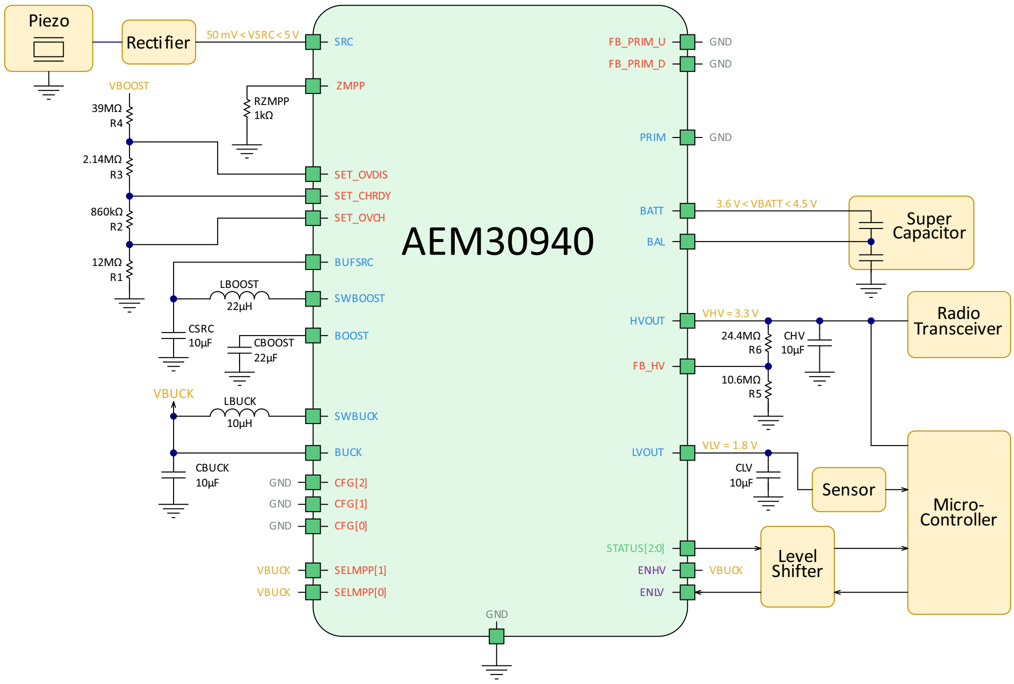


Figure 18: Typical application circuit 2

The energy source is a piezo source and the storage element is a dual-cell supercapacitor. Please note that the supercapacitor might be completely depleted during the cold start.

Moreover, BAL is connected to the dual-cell supercapacitor to compensate for any mismatch between the two cells and, in that way, protect the supercapacitor.

A micro-controller acts as the application master. The operating mode pins are set as follows:

- CFG[2:0] = LLL (all to GND)

The storage element voltages are set as follows with a custom configuration:

- V_{OVCH} = 4.50 V
- V_{CHRDY} = 4.20 V
- V_{OVDIS} = 3.60 V

R_T is set to 54 MΩ. R1, R2, R3 and R4 values are computed from the equations in Section 6.1.1:

- $R1 = 54M\Omega \cdot \frac{1V}{4.5V} = 12M\Omega$
- $R2 = 54M\Omega \cdot \left(\frac{1V}{4.2V} - \frac{1V}{4.5V}\right) = 860k\Omega$
- $R3 = 54M\Omega \cdot \left(\frac{1V}{3.6V} - \frac{1V}{4.2V}\right) = 2.14M\Omega$
- $R4 = 54M\Omega \cdot \left(1 - \frac{1V}{3.6V}\right) = 39M\Omega$

The LDO voltages are set as follows:

- V_{HV} = 3.3 V
- V_{LV} = 1.8 V

LVOUT output enabling is controlled by the application circuit with a micro-controller GPIO connected to ENLV.

ENHV is tied to V_{BUCK} so that HVOUT is always on.

R_V is set to 35 MΩ. R5 and R6 are determined by applying the equations found in Section 6.1:

- $R5 = 35M\Omega \cdot \frac{1V}{3.3V} = 10.6M\Omega$
- $R6 = 35M\Omega \cdot \left(1 - \frac{1V}{3.3V}\right) = 24.4M\Omega$

The micro-controller is supplied by HVOUT, which is enabled when V_{BATT} and V_{BOOST} voltages rise above V_{CHRDY}.

The MPPT configuration pins SELMPP[1:0] are tied to V_{BUCK} (logic HIGH), thus, selecting the ZMPPT configuration to match a 1 kΩ impedance..

No primary battery is connected: PRIM, FB_PRIM_U and FB_PRIM_D pins are tied to GND.

8. Circuit Behavior

8.1. Cold-start Behavior with a (Super)Capacitor as Storage Element

8.1.1. Configuration

- CFG[2:0] = LHL
  - V_{OVCH} = 4.50 V
  - V_{CHRDY} = 3.92 V
  - V_{OVDIS} = 3.60 V
  - V_{HV} = 3.3 V
  - V_{LV} = 1.8 V
- C_{BATT} = 4.85 mF
- SELMPP[1:0] = HL
  - V_{MPP}/V_{OC} ratio = 80%
- SRC: 1 mA current source with 3 V voltage compliance
  - V_{OC} = 3.0 V
  - V_{MPP} = 2.4 V
  - I_{SRC} = 1 mA
- ENHV = ENLV = H
- 22 kΩ resistive load on LVOUT
- 22 kΩ resistive load on HVOUT

8.1.2. Observations

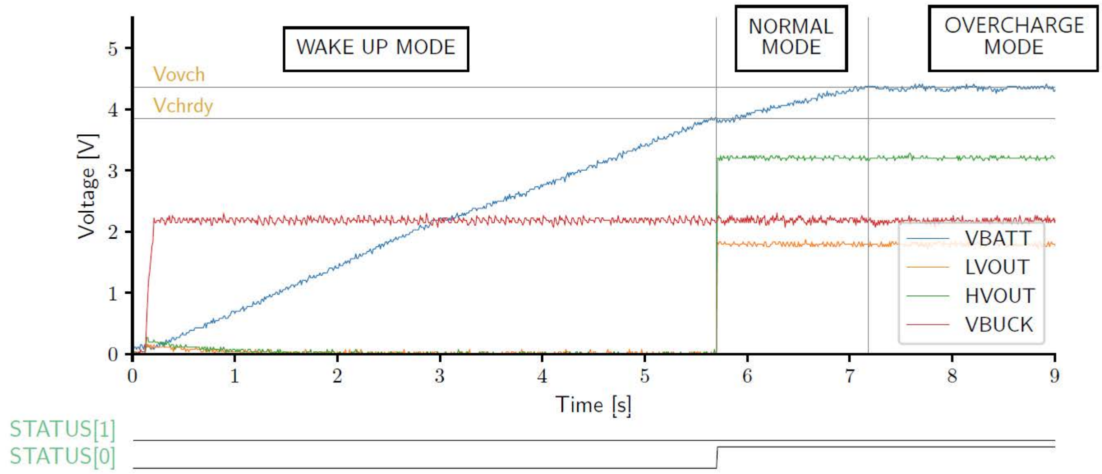


Figure 19: Cold start with a capacitor connected to BATT

- The AEM30940 is initially in DEEP SLEEP MODE with all nodes fully discharged. As soon as the current source is connected on SRC, the AEM30940 switches to WAKE-UP MODE, V_{BUCK} rises to its regulation voltage (around 2.2 V), and the storage element connected on BATT starts to be charged.
- Once V_{BATT} reaches V_{CHRDY}, the AEM30940 switches to NORMAL MODE, STATUS[0] is asserted and the LDOs are enabled. V_{HV} and V_{LV} are respectively regulated to 3.3 V and 1.8 V.
- Once V_{BATT} reaches V_{OVCH}, the AEM30940 switches to OVERCHARGE MODE. The voltage on BATT is maintained around V_{OVCH}. V_{BUCK}, V_{HV} and V_{LV} remain regulated.

8.2. Cold-start Behavior with a Battery as Storage Element

8.2.1. Configuration

- CFG[2:0] = HHH
  - V_{OVCH} = 4.12 V
  - V_{CHRDY} = 3.67 V
  - V_{OVDIS} = 3.60 V
  - V_{HV} = 3.3 V
  - V_{LV} = 1.8 V
- C_{BATT} = 4.85 mF (pre-charged capacitor acting as a battery
- SELMPP[1:0] = HL
  - V_{MPP}/V_{OC} ratio = 80%
- SRC: 1 mA current source with 3 V voltage compliance
  - V_{OC} = 3.0 V
  - V_{MPP} = 2.4 V
  - I_{SRC} = 1 mA
- ENHV = ENLV = H
- 22 kΩ resistive load on LVOUT
- 22 kΩ resistive load on HVOUT

8.2.2. Observations

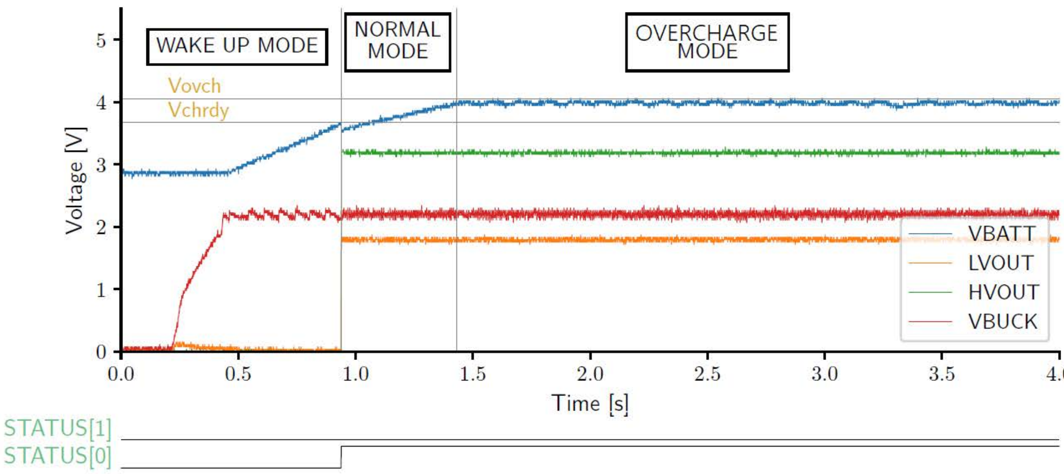


Figure 20: Cold start with a battery connected to BATT

- The AEM30940 is initially in DEEP SLEEP MODE with all nodes fully discharged except V_{BATT} which is pre-charged. As soon as the current source is connected on SRC, the AEM30940 switches to WAKE-UP MODE, V_{BUCK} rises to its regulation voltage (around 2.2 V), and the storage element connected on BATT starts to be charged.
- Once V_{BATT} reaches V_{CHRDY}, the AEM30940 switches to NORMAL MODE, STATUS[0] is asserted and the LDOs are enabled. V_{HV} and V_{LV} are respectively regulated to 3.3 V and 1.8 V.
- Once V_{BATT} reaches V_{OVCH}, the AEM30940 switches to OVERCHARGE MODE. The voltage on BATT is maintained around V_{OVCH}. V_{BUCK}, V_{HV} and V_{LV} remain regulated.

8.3. Overcharge Mode Behavior

8.3.1. Configuration

- CFG[2:0] = HHH
  - V_{OVCH} = 4.12 V
  - V_{CHRDY} = 3.67 V
  - V_{OVDIS} = 3.60 V
  - V_{HV} = 3.3 V
  - V_{LV} = 1.8 V
- C_{BATT} = 4.85 mF
- SELMPP[1:0] = HL
  - V_{MPP}/V_{OC} ratio = 80%
- SRC: 1 mA current source with 3 V voltage compliance
  - V_{OC} = 3.0 V
  - V_{MPP} = 2.4 V
  - I_{SRC} = 1 mA
- ENHV = ENLV = H
- 22 kΩ resistive load on LVOUT
- 22 kΩ resistive load on HVOUT

8.3.2. Observations

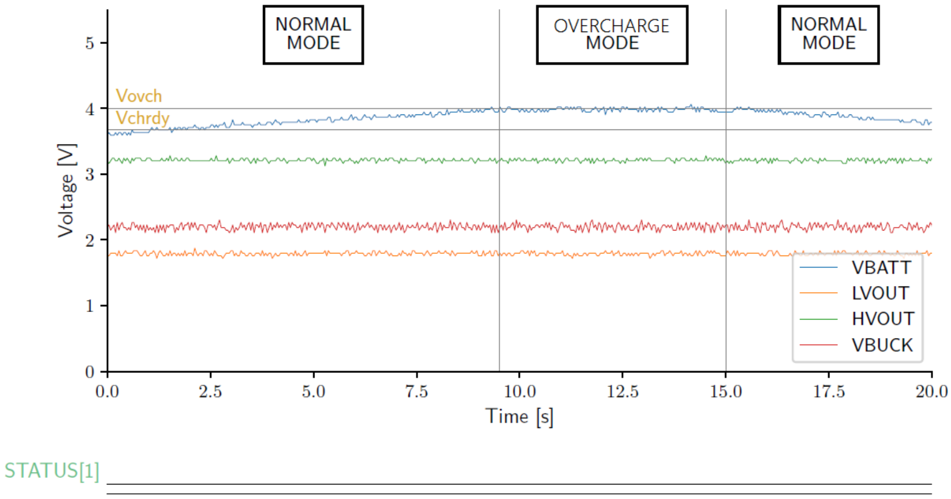


Figure 21: Overcharge mode

- The AEM30940 is initially in NORMAL MODE, with STATUS[0] asserted, and V_{BUCK}, V_{HV} and V_{LV} regulated.
- Thanks to the current source connected on SRC, V_{BATT} is being charged.
- Once V_{BATT} reaches V_{OVCH}, the AEM30940 enters OVERCHARGE MODE and maintain V_{BATT} around V_{OVCH} to avoid damaging the storage element.
- Near 15 s, the current source is disconnected from SRC. The AEM30940 maintain the two LDOs supplied, causing the storage element to discharge. The AEM30940 switches back to NORMAL MODE.

8.4. Shutdown Mode Behavior

8.4.1. Configuration

- CFG[2:0] = LHL
  - V_{OVCH} = 4.50 V
  - V_{CHRDY} = 3.92 V
  - V_{OVDIS} = 3.60 V
  - V_{HV} = 3.3 V
  - V_{LV} = 1.8 V
- C_{BATT} = 4.85 mF
- SELMPP[1:0] = HL
  - V_{MPP}/V_{OC} ratio = 80%
- SRC: left floating to let the storage element on BATT discharge
- ENHV = ENLV = H
- 22 kΩ resistive load on LVOUT
- 22 kΩ resistive load on HVOUT
- PRIM, FB_PRIM_U and FB_PRIM_D connected to GND (no primary battery)

8.4.2. Observations

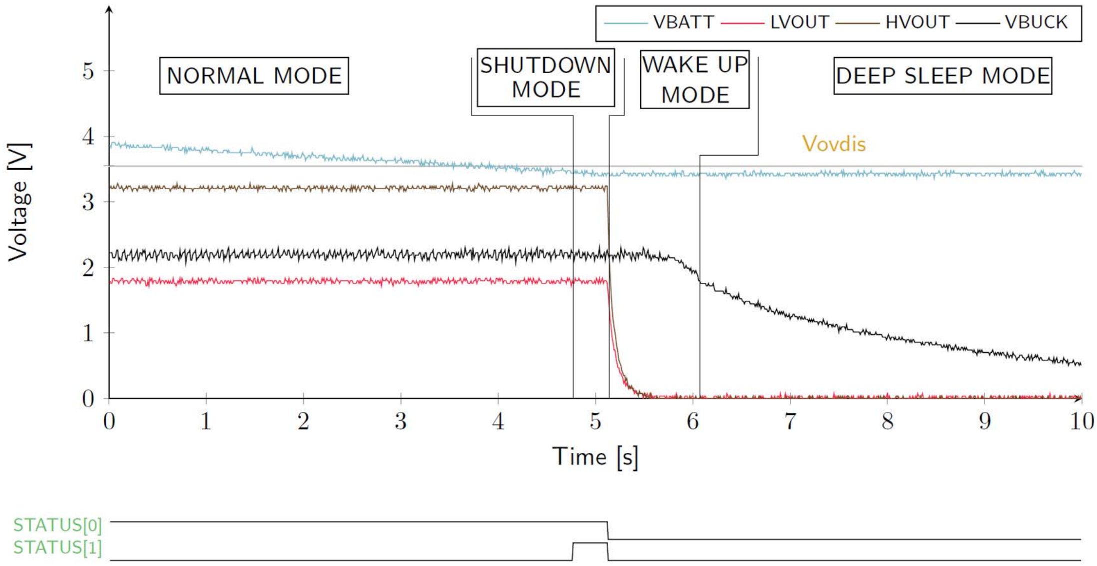


Figure 22: Shutdown mode (without primary battery)

- The AEM30940 is initially in NORMAL MODE, with STATUS[0] asserted, V_{BUCK}, V_{HV} and V_{LV} regulated. The SRC pin is left floating, which result in the storage element discharging through the loads.
- Once V_{BATT} reaches V_{OVDIS}, the AEM30940 enters SHUTDOWN MODE and maintain the load outputs regulated. STATUS[1] is asserted to warn the user that a shutdown may occur.
- After T_{CRIT} (approximately 600 ms), V_{BATT} has not recovered above V_{OVDIS}. The AEM30940 disables the LDOs and enters WAKE-UP MODE. To avoid further discharging the storage element, BATT is disconnected from BOOST. The pins STATUS[1] and STATUS[0] are set LOW.
- Near 6.1 s, V_{BUCK} falls below $V_{BUCK,RESET}$. The AEM30940 enters DEEP SLEEP MODE and must go through the cold-start procedure again.

8.5. Primary Battery Mode Behavior

8.5.1. Configuration

- CFG[2:0] = HHH
  - V_{OVCH} = 4.12 V
  - V_{CHRDY} = 3.67 V
  - V_{OVDIS} = 3.60 V
  - V_{HV} = 3.3 V
  - V_{LV} = 1.8 V
- C_{BATT} = 4.85 mF
- SELMPP[1:0] = HL
  - V_{MPP}/V_{OC} ratio = 80%
- SRC: 1 mA current source with 3 V voltage compliance (initially disconnected)
  - V_{OC} = 3.0 V
  - V_{MPP} = 2.4 V
  - I_{SRC} = 1 mA
- ENHV = ENLV = H
- 22 kΩ resistive load on LVOUT
- 22 kΩ resistive load on HVOUT
- PRIM: 3 V voltage source with 1 mA current compliance
  - R7 = 68 kΩ
  - R8 = 330 kΩ
  - $V_{PRIM,MIN}$ = 1.50 V

8.5.2. Observations

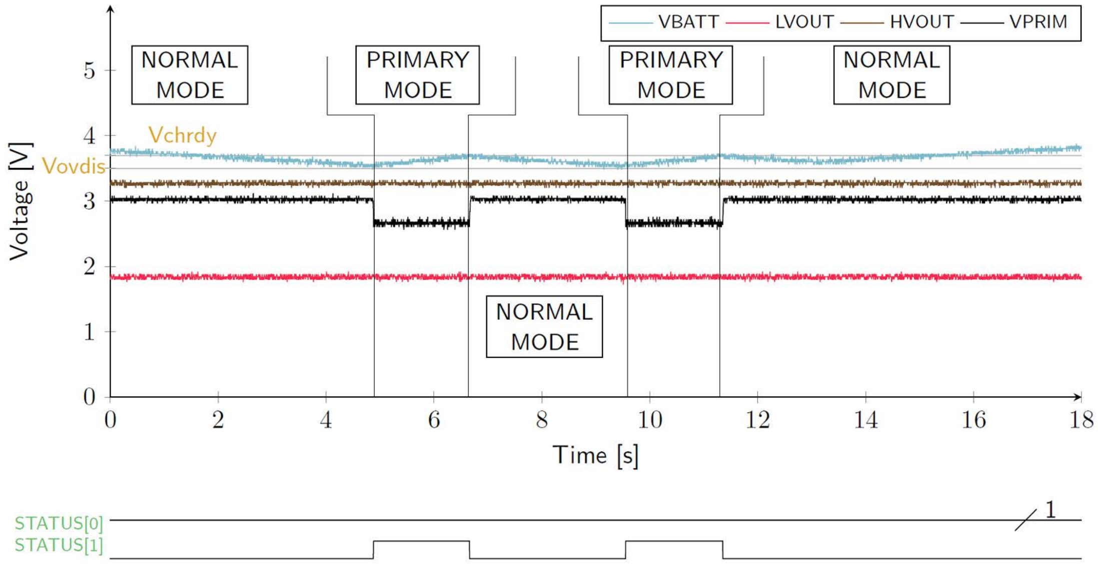


Figure 23: Switching to primary battery when battery is overdischarged

- The AEM30940 is initially in NORMAL MODE, with STATUS[0] asserted, V_{BUCK}, V_{HV} and V_{LV} regulated.
- The SRC is left floating, causing the storage element to discharge.
- Once V_{BATT} reaches V_{OVDIS}, the AEM30940 compares the voltage on PRIM to $V_{PRIM,MIN}$. Since V_{PRIM} is higher, the AEM enters PRIMARY BATTERY MODE and, the primary battery is used to recharge the storage element until V_{BATT} reaches V_{CHRDY} again.
- STATUS[1] is asserted as long as the AEM30940 is in PRIMARY BATTERY MODE.
- Once V_{BATT} reaches V_{CHRDY}, the AEM30940 goes back into NORMAL MODE. Since the SRC is still floating, the same behavior repeats.
- Near 13 s, while the storage element is discharging in NORMAL MODE, the current source is connected on SRC. This allows charging the storage element so the primary battery is not used anymore.

9. Schematic

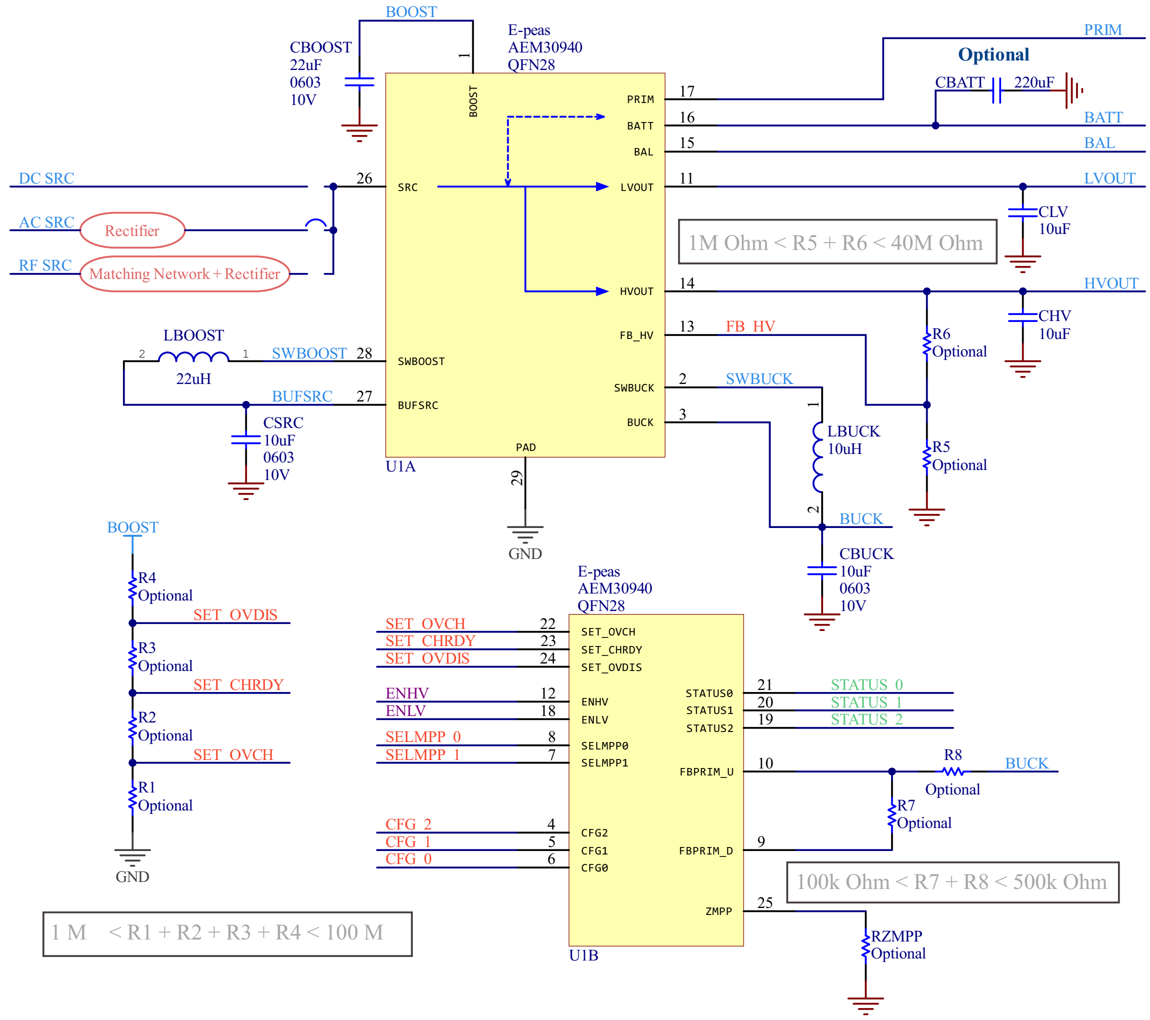


Figure 24: Schematic example

Designator	Description	Qty	Manufact.	Part Number
U1	AEM30940, QFN28	1	e-peas	order at sales@e-peas.com
C_{SRC}	Ceramic Cap 10 µF, 16 V, 20 %, X5R, 0603	1	Murata	GRM188R61C106MA
C_{BOOST}	Ceramic Cap 22 µF, 10 V, 20 %, X5R, 0603	1	Murata	GRM188R61A226ME
L_{BOOST}	Power Inductor 22 µH, 0.65 A, LPS4018	1	Coilcraft	LPS4018-223MR
L_{BOOST} (alt.)	Power Inductor 10 µH, 0.90 A, LPS4018	1	Coilcraft	LPS4018-103MR
L_{BOOST} (alt.)	Power Inductor 10 µH, 0.84 A, 3015	1	Würth	744 040 321 00
C_{BUCK}	Ceramic Cap 10 µF, 16 V, 20 %, X5R, 0603	1	Murata	GRM188R61C106MA
L_{BUCK}	Power Inductor 10 µH, 0.25 A, 0603	1	TDK	MLZ1608N100LT000
C_{LV}	Ceramic Cap 10 µF, 16 V, 20 %, X5R, 0603	1	Murata	GRM188R61C106MA
C_{HV}	Ceramic Cap 10 µF, 16 V, 20 %, X5R, 0603	1	Murata	GRM188R61C106MA
C_{BATT}	Ceramic Cap 220 µF, 6.3 V, 20 %, X5R, 1206	1	Murata	GRM31CR60J227ME

Table 11: BOM example for AEM30940 and its required passive components

10. Layout

10.1. Guidelines

Good layout practices are mandatory in order to obtain good AEM30940 stability, best efficiency and avoid EMI problems.

The following list, while not exhaustive, shows the main attention points when routing a PCB with the AEM30940:

- The switching nodes (SWBOOST and SWBUCK) must be kept as short as possible, with minimal track resistance and minimal track capacitance. Low resistance is obtained by keeping track length as short as possible and track width as large as possible between these switching nodes and the AEM30940 pins. Minimal capacitance is obtained by maintaining a large distance between the switching nodes and other signals. We recommend removing the ground plane, the power plane and the bottom layer ground pour under L_{BOOST} and L_{BUCK} footprints, as well as adding distance between BUFSRC/SWBOOST and the top ground pour, as shown in Figure 25.
- The decoupling capacitors (C_{BOOST} - C_{BUCK} - C_{SRC} - C_{HV} - C_{LV} - C_{BATT}) must be placed as close as possible to the AEM30940, with direct connection and minimum track resistance for the corresponding power nodes (BOOST, BUCK, BUFSRC, HVOUT, LVOUT and BATT).
- The GND return path between the decoupling capacitors and the AEM30940 thermal pad, which is the AEM30940 main GND connection, must be as direct and short as possible. This is preferably done on the top layer when possible, otherwise by internal/bottom plane, using low resistance vias to decrease layer-to-layer connection resistance. In Figure 25, this GND return path is done on an internal plane.
- The external DC power connections (SRC, HVOUT, LVOUT and BATT) must be connected to the AEM30940 with low resistance tracks.
- If used, ZMPP must be connected to the AEM30940 with a low resistance track, according to the expected SRC power.
- The BAL pin connection track must be able to handle at least 40 mA.
- The custom mode setting pins SET_OVDIS, SET_CHRDY and SET_OVCH are high impedance analog inputs typically connected to a resistive divider with high resistor values, making those three nodes prone to pickup noise. Thus, it is recommended to keep those as short as possible and as far as possible to noise sources such as DCDC switching nodes.
- The configuration pins and the status pins have minimal layout restrictions.

10.2. Layout Example

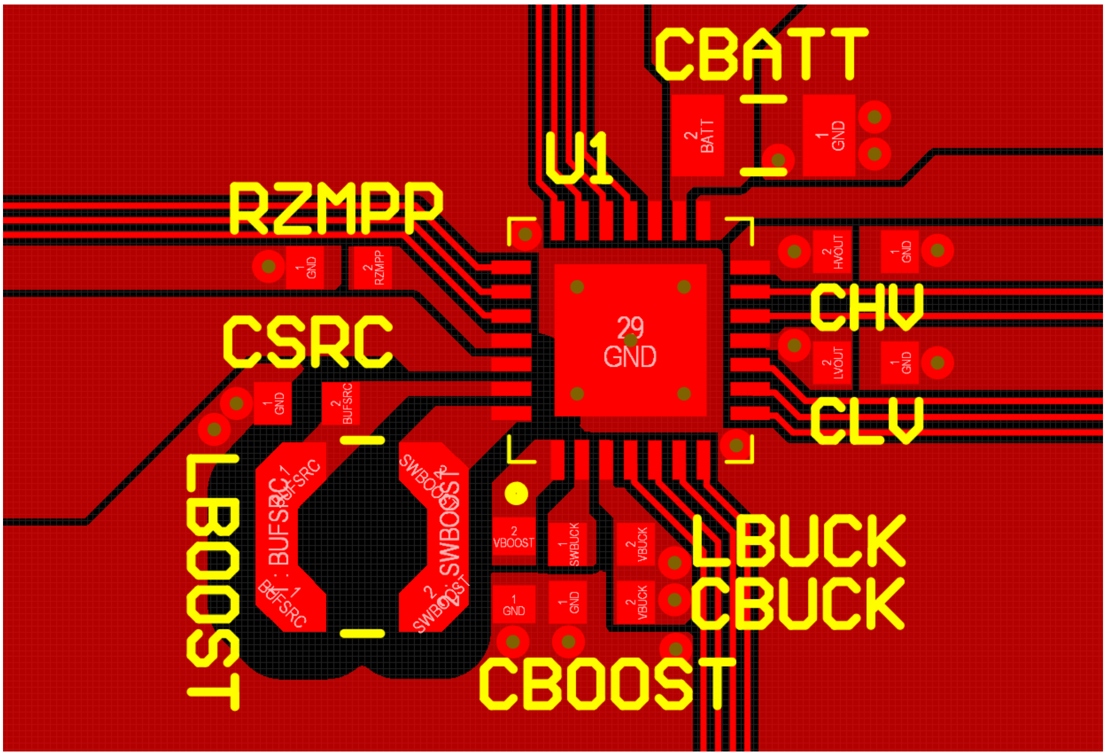


Figure 25: Layout example for the AEM30940 and its passive components

11. Package Information

11.1. Moisture Sensitivity Level

Package	Parameter	Value
QFN 28-pin 5x5mm	Moisture Sensitivity Level (MSL)[1]	Level 1

Table 12: Moisture sensitivity level

1. According to JEDEC 22-A113 standard.

11.2. RoHS Compliance

e-peas product complies with RoHS requirement.

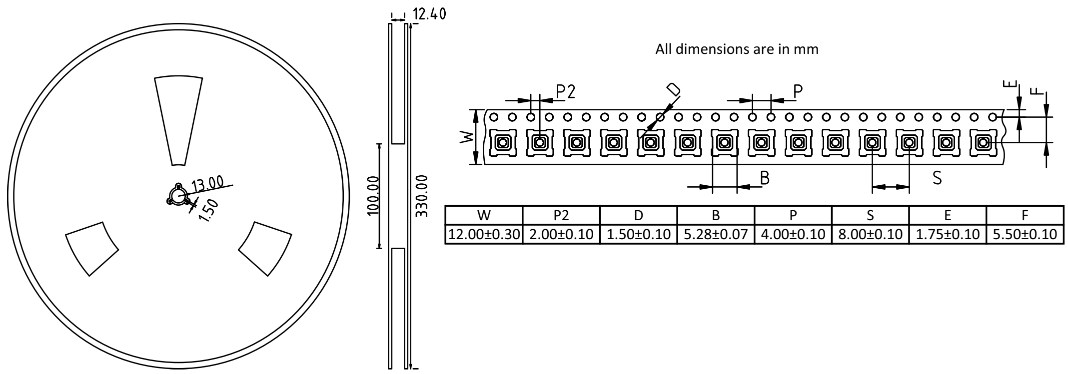
e-peas defines “RoHS” to mean that semiconductor end-products are compliant with RoHS regulation for all 10 RoHS substances.

This applies to silicon, die attached adhesive, gold wire bonding, lead frames, mold compound, and lead finish (pure tin).

11.3. REACH Compliance

The component and elements used by e-peas subcontractors to manufacture e-peas PMICs and devices are REACH compliant. For more detailed information, please contact e-peas sales team.

11.4. Tape and Reel Dimensions



W	P2	D	B	P	S	E	F
12.00±0.30	2.00±0.10	1.50±0.10	5.28±0.07	4.00±0.10	8.00±0.10	1.75±0.10	5.50±0.10

Figure 26: Tape and reel dimensions

11.5. Package Dimensions

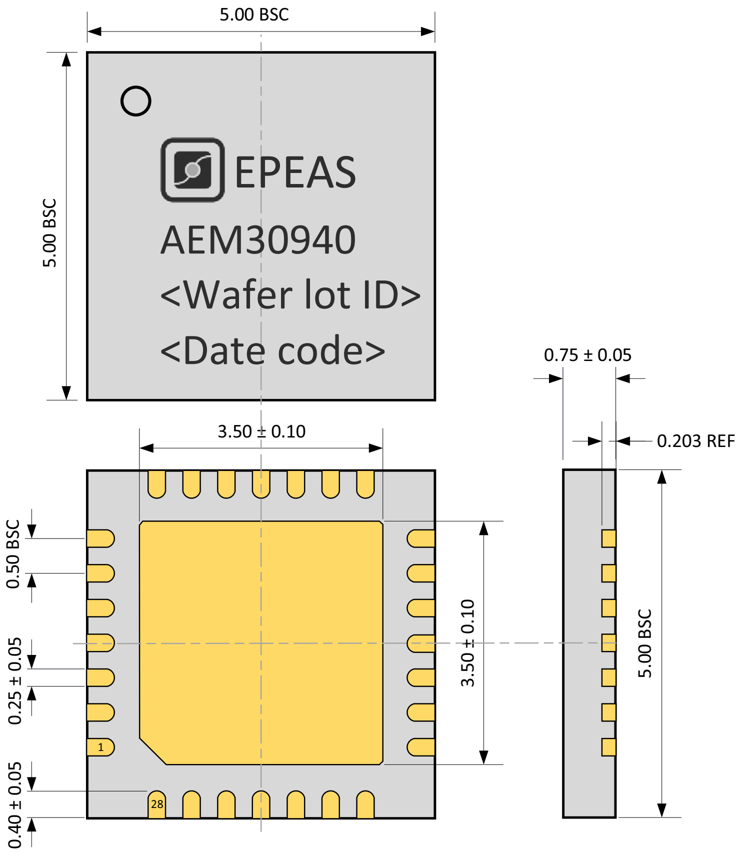


Figure 27: QFN 28-pin 5x5mm drawing (all dimension in mm)

11.6. Board Layout

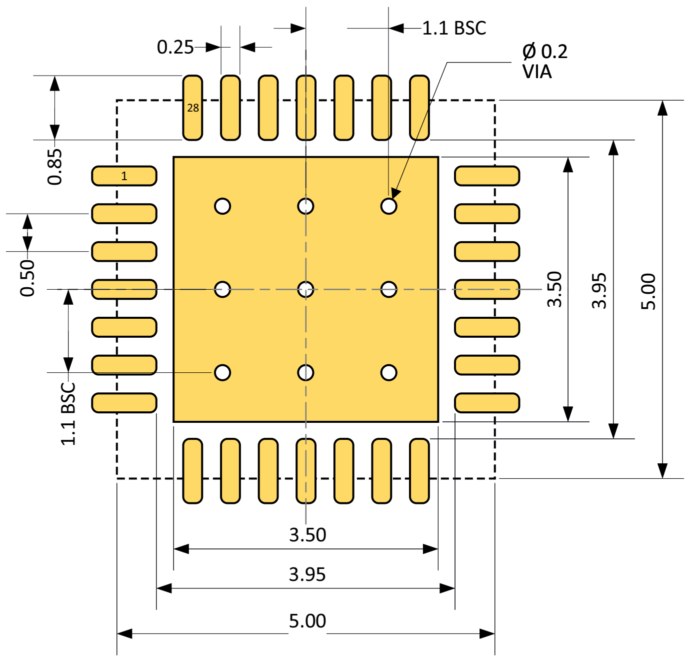


Figure 28: Recommended board layout for QFN 28-pin 5x5mm (all dimension in mm)

12. Glossary

Symbol	Description
AEM	Ambient Energy Manager.
BOM	Bill Of Materials.
C_{BATT}	Capacitor connected on the BATT pin.
C_{BOOST}	Output capacitor of the BOOST converter.
C_{BUCK}	Output capacitor of the BUCK converter.
C_{HV}	High-voltage LDO regulator decoupling capacitor.
C_{LV}	Low-voltage LDO regulator decoupling capacitor.
C_{SRC}	BUFSRC pin decoupling capacitor.
GPIO	General Purpose Input / Output.
I_{BUCK}	Total load current supplied by the BUCK converter (including the LVOUT current I_{LV}).
I_{HV}	Load current supplied by the high-voltage LDO regulator.
I_{LV}	Load current supplied by the low-voltage LDO regulator.
I_{PRIM}	Current from the primary battery.
I_Q	Quiescent current on BATT when no energy is available on SRC.
I_{SRC}	Harvested current from the energy source.
L_{BOOST}	BOOST converter inductor.
L_{BUCK}	BUCK converter inductor.
LDO	Low Drop-Out linear regulator.
MPPT	Maximum Power Point Tracking.
PCB	Printed Circuit Board.
$P_{SRC,CS}$	Minimum power on SRC for the AEM30940 to coldstart.
R_P	Sum of resistors for setting the primary battery minimum voltage. R_P = R7 + R8.
R_T	Sum of resistors for setting the battery protection threshold voltages in custom mode. R_T = R1 + R2 + R3 + R4.
R_V	Sum of resistors for setting the output voltage of the high-voltage LDO in custom mode. R_V = R5 + R6.
R_{ZMPP}	Resistor that defines the AEM30940 BOOST converter input resistance when used in ZMPP mode.
T_{CRIT}	Time during which the AEM stays in SHUTDOWN MODE before returning to NORMAL MODE or going to DEEP SLEEP MODE.
$T_{MPPT,MEASURE}$	Duration of V_{OC} measurement during MPP evaluations.
$T_{MPPT,PERIOD}$	Time between two MPP evaluations.
$T_{MPPT,WAIT}$	Time during which the AEM30940 stops pulling current on SRC before measuring the harvester open circuit voltage (V_{OC}).
V_{BAL}	Voltage on the BAL pin.
$V_{BAL,MIN}$	Minimum voltage on the BAL pin for the balancing circuit to operate.
V_{BATT}	Voltage on the BATT pin.
V_{BOOST}	Output voltage of the BOOST converter.
V_{BUCK}	Output voltage of the BUCK converter.
$V_{BUCK,RESET}$	Minimum voltage on BUCK before switching to DEEP SLEEP MODE (from any other mode).

Table 13: Glossary (Part 1)

Symbol	Description
V_{CHRDY}	Charge ready voltage on the BATT pin.
$V_{FB\_PRIM\_U}$	Feedback for the minimal voltage level on the primary battery.
V_{HV}	Output voltage of the high-voltage LDO regulator.
V_{LV}	Output voltage of the low-voltage LDO regulator.
V_{MPP}	Target regulation voltage on SRC when extracting power.
V_{OC}	Open-circuit voltage of the harvester connected to the SRC pin.
V_{OVCH}	Over-charge voltage on the BATT pin.
V_{OVDIS}	Over-discharge voltage on the BATT pin.
V_{PRIM}	Voltage on the primary battery.
$V_{PRIM,MIN}$	Voltage at which the primary battery is considered fully depleted.
V_{SRC}	Voltage on the SRC pin.
$V_{SRC,CS}$	Minimum voltage on SRC for the AEM30940 to coldstart.
ZMPPT	Maximum Power Point Tracking with constant impedance.

Table 13: Glossary (Part 2)

13. Revision History

Revision	Date	Description

1.7	April, 2026	- Document structure updated (section order and titles revised).
		- Updated HBM ESD value.
		- Addition of REACH and RoHS compliance statements.
		- VPRIM maximum value modified for VOVDIS + 0.4 V. Corresponding explanations added in the Boost Converter section.
		- VBAL,MIN value added in Electrical Characteristics table.
		- Storage Element Balancing Circuit section reworked.
		- Updated the MPP evaluation behavior figure.
		- Updated terminology for MPPT timing parameters throughout the document for improved clarity and consistency:
		    - $T_{MPPT,VOC}$ modified to $T_{MPPT,WAIT}$.
		    - $T_{MPPT,MEASURE}$ added.
		- Addition of a disclaimer about the battery threshold levels.
		- Reworked and moved the external inductors, external capacitors, and storage element information sections.
		- Circuit Behavior sections rework.
		- Change LBOOST recommenced value to 22 µH for best efficiency throughout the datasheet.
		- Change CBATT recommended value to 220 µF in “Schematic” section.
		- Updated LBUCK part number in the BOM example table.
		- Updated VPRIM maximum value in the Simplified schematic view and Typical application circuit 1 figures.
		- Added moisture sensitivity level and tape and reel dimensions in the package information section.
		- Updated the package dimensions and the recommended board layout.

1.6	March, 2024	- STATUS[1] description modified in pin description table. - VBUCK,RESET value added in Electrical Characteristics table. - Diagram of the AEM30940 modes modified with new links between modes. - Primary Battery Mode section and Shutdown Mode section reworked. - Primary battery mode behavior graph moved into a dedicated section. - Efficiency graphs replaced with new ones.

Table 14: Revision history (Part 1)

Revision	Date	Description

1.5	December, 2023	- RF input power values and efficiency graphs removed and placed in an app note.
		- First page:
		  - Maximum MPPT voltage operation range corrected to 4.5 V.
		  - SET_OVCH and SET_OVDIS swapped on AEM diagram.
		- Added all AEM pins in the Absolute Maximum Ratings table.
		- Added “Pin Configuration and Functions” section with pinout diagram and functions.
		- SET_OVCH, SET_CHRDY and SET_OVDIS pins changed from “left floating” to “connected to BUCK” when not used in Table 1 and in Section 6.1.
		- Voltage on BATT and PRIM added in Absolute Maximum Ratings table.
		- Table 6:
		  - VMPP minimum and maximum values added.
		  - ISRC value added for LBOOST value of 22 µH.
		  - Different min values for VBATT depending on used storage element removed.
		  - VHV maximum value changed from “VBATT - 0.3 V” to “VOVDIS - 0.3 V”.
		  - Added IBUCK, IQ, TMPPT.VOC and TMPPT.PERIOD.
		- Table 7:
		  - CSRC maximum value changed from 150 µF to 22 µF.
		  - LBOOST, CBOOST and CBUCK maximum values removed.
		  - CBUCK min and typical values modified from 15 µF and 22 µF to 8 µF and 10 µF.
		  - ENHV minimum and maximum values removed.
		  - ENLV minimum value replaced by VBUCK.
		  - Specified CBATT minimum value if LDOs enabled or disabled.
		- Condition to go from SHUTDOWN mode to NORMAL mode corrected to “If VBATT > VOVDIS” in Figure 13 and Section 5.2.5.
		- Maximum Power Point Tracking section rephrased and MPP behavior figure added.
		- Added how to use BUCK to supply an application, and BUCK converter performances.
		- CBATT of 22 µF added when no storage element and LDOs not used (Section 6.8).
		- Second LBOOST recommended value of 22 µH added in Section 6.9 and Table 11.
		- SET_OVCH, SET_CHRDY and SET_OVDIS pins connected to VBUCK in Figure 2.
		- Modified VOVDIS, R2, R3 and R4 values in Example Circuit 2.
		- Replaced LBOOST LPS4012-103MR by LPS4018-103MR in BOM example table.
		- New section created to place the circuit behavior figures.
		- Layout example replaced by a more recent version.
		- Added layout guidelines section and glossary section.
		- Replaced 0 by L for logic LOW and 1 by H for logic HIGH in tables and texts.

1.4	June, 2022	CBUCK value changed from 10 µF to 22 µF.
1.3	November, 2020	- Performances efficiency curves added for the 2450 MHz frequency band. - ESD qualification added.
1.2	June, 2019	- Efficiencies measurements updated. - ESD specifications. - HVOUT voltage changed from 4.2 V to 4.1 V in Figure 3. - AC source instead of Power receiving antenna for the front page Figure and the Figures 4, 11 and 12.
1.0	July, 2018	Creation of the document.

Table 14: Revision history (Part 2)